5 Pulse Modulation Schemes

Objective

After completing this chapter, user will be able to understand

- Nyquist Rate and Nyquist interval
- What is sampling and its types?
- Pulse analog modulation and its types
- Pulse digital modulation and its types

Introduction

In continuous-wave modulation, some parameters of carrier wave are varied continuously with respect to message signal. This is completely different in the case of pulse modulation, which will be discussed later in this chapter.

In pulse modulation, some parameters of a pulse train are varied with respect to the message signal. We can differentiate pulse modulation as analog pulse modulation and digital pulse modulation. In analog pulse modulation, a periodic pulse train serves as the carrier wave, and some characteristic attribute of each pulse, such as amplitude, duration or location, is continuously modified with respect to message signal. Thus, in analog pulse modulation, information is transmitted in analog form, but the transmission takes place at discrete times.

While in digital pulse modulation, the message signal is denoted as a signal which is discrete in time and amplitude, thereby permitting its transmission in digital form as a sequence of coded pulses.

Sampling

In most cases, a sampling process is given in the time domain. It is a fundamental operation in digital signal processing and communication. With the help of the sampling process, an analog signal is converted into a corresponding sequence of samples that are usually spaced uniformly in time. A common use of sampling is for converting a continuous-time signal to an equivalent discrete time signal and vice versa. The concept and techniques of sampling a continuous-time signal have important roles in base-band signal processing like digitization, multiplexing, filtering and also in carrier modulations in wireless digital communication systems. The Nyquist theorem is the basic sampling theorem for preprocessing continuous-time signals.

Sampling Theorem

The Sampling theorem states that a continuous signal band limited to f_m Hz can be reconstructed from the sampled signal if and only if

$F_s \geq 2f_m$ samples/sec

$$T_s \geq \frac{1}{2f_m} \text{sec}$$

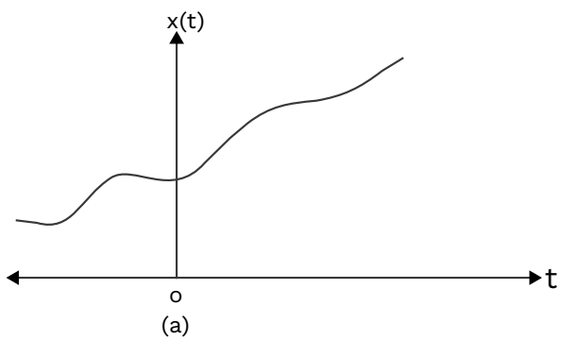
where

T_s = Sampling interval (sec)

f_s = Sampling rate or sampling frequency

- Depending on the sampling rate, sampling is divided as
  - $f_s \leq 2f_m \rightarrow$ Under sampling
  - $f_s > 2f_m \rightarrow$ Over sampling
  - $f_s = 2f_m \rightarrow$ Critical sampling

Proof of sampling theorem:



(a)

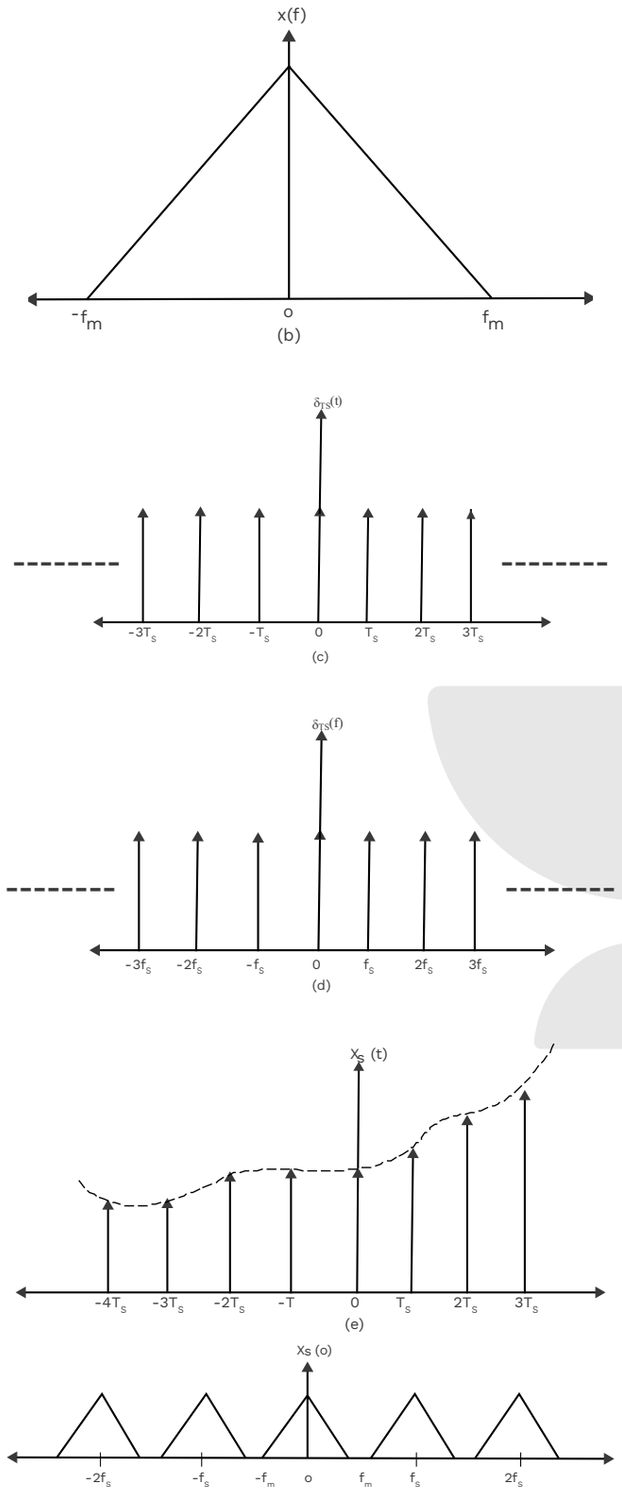


Fig. 5.1 (A) Message Signal (B) Spectrum of Message Signal (C) Impulse Train (D) Spectrum of Impulse Train (E) Sampled Signal (F) Sample Signal Spectrum.

$x(t) \xrightarrow{\text{Fourier transform}} x(f)$

$$x_s(t) = x(t).\delta_{T_s}(t)$$

$$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$x_s(t) = x(t).\sum_{-\infty}^{\infty} \delta(t - nT_s)$$

$$\delta_s(t) = \frac{1}{T_s}\left[1 + 2\cos f_s t + 2\cos f_s t + 2\cos 3f_s t + ...\right]$$

Now,

$$X_s(t) = \left[x(t) + 2x(t)\cos f_s t + 2x(t)\cos 2f_s t....\right]$$

$X(t) \longleftrightarrow x(f)$

$X_s(t) \longleftrightarrow X_s(f)$

$2x(t)\cos f_s t \longleftrightarrow X(f - f_s) + (f + f_s)$

(Frequency shifting property)

$$X_s(f) = \frac{1}{T_s}$$

$$\left[X(f) + X(f - f_s) + X(f + f_s) + X(f + 2f_s) + X(f - 2f_s).....\right]$$

$$X_s(f) = f_s \sum_{n=-\infty}^{\infty} X(f - nf_s)$$

Now, if we want to reconstruct x(t) from $X_s(t)$, we must be able to recover X(f) from $X_s(f)$ and this is possible only if there is no overlap between consecutive cycles of $X_s(f)$ as shown in Figure 4.1(f). This can happen if and only if

$$f_s \geq 2f_m$$

Therefore, if the sampling frequency f_s is greater than twice the maximum signal frequency f_m, $X_s(f)$ will consist of non-overlapping repetition.

Nyquist Rate and Nyquist Interval

The minimum sampling rate and maximum sampling interval to be maintained to avoid ALIASING are known as “Nyquist Rate” and “Nyquist Interval” respectively.

Nyquist Rate = $2f_m$ Hz

Nyquist Interval = $\frac{1}{2f_m}$ sec

Aliasing or fold over

When the sampling frequency is less than the Nyquist frequency, the original signal is not recoverable by low pass filtering because sampled signals overlap in the frequency domain, and this condition is referred to as aliasing.

To avoid aliasing, we have to band-limit the signal by passing through a filter (called an Anti-aliasing filter).

Types of Sampling

Based on sampling operation:

- Instantaneous sampling or ideal sampling
- Natural sampling
- Flat-top sampling

Based on the signal spectrum:

- Base-band sampling
- Band-pass sampling

$X_s(t)$ is spectrum of X(t), sampled with uniform sampling rate "$\frac{1}{T_s}$"

$$X_s(f) = \sum_{n=-\infty}^{\infty} C_n X(f - nf_s)$$

$C_n = \frac{1}{T_s}$ for Instantaneous sampling T_s

$C_n = \frac{A\tau}{T_s}\text{sinc}(nf_s\tau)$ for natural sampling T_s

- Instantaneous sampling is not practically realizable since it is impossible to generate true impulses, and the spectrum of such an ideal sampled signal occupies the entire bandwidth (-∞, ∞) i.e., it contains components of all frequencies. Instantaneous sampling uses the multiplication of the impulse train with the signal.
- In natural sampling, the tops of the sampled signal follow the natural envelope of the message signal; hence it is known as natural sampling. It is done by using the chopper circuit.

$$X_s(f) = \frac{\tau A}{T_s}\sum_{n=-\infty}^{\infty}\text{sinc}(nf_s\tau)X(f - nf_s).$$

- In Flat top sampling, the tops of the sampled signal are flat. It is done by using a sample and hold circuit.

$$X_s(f) = f_s\sum_{n=-\infty}^{\infty}X(f - nf_s)H(f)$$

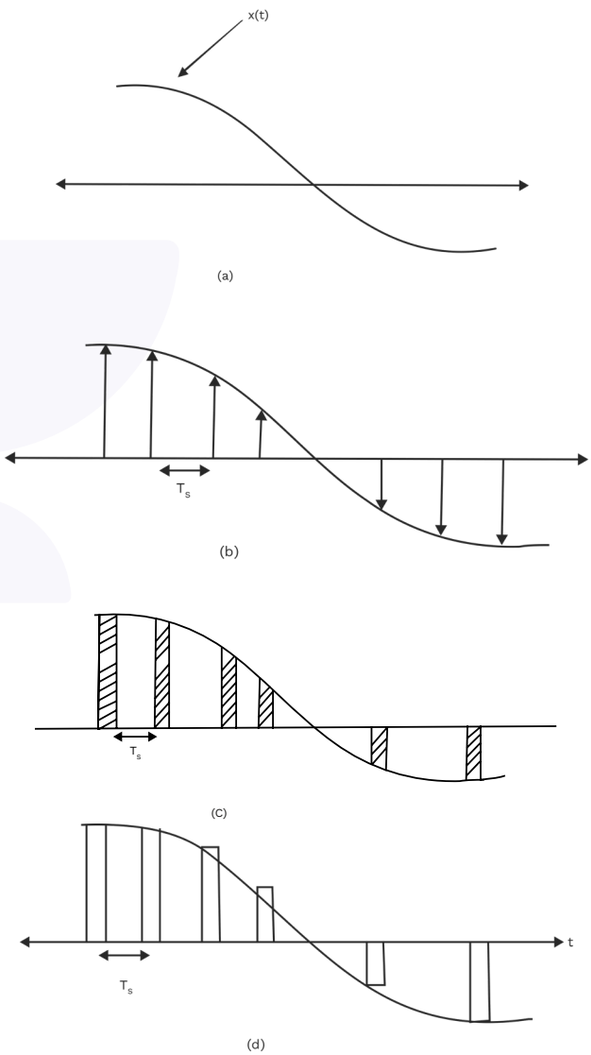


Fig. 5.2 (A) Modulation Signal (B) Ideal Sampled Signal (C) Natural Sampled Signal (D) Flat-top Sampled Signal

SUMMARY

1. $F_s > 2F_m$: Oversampling
   Recovery:
   $(f_s > 2f_m) \cap (f_m < f_c < f_s - f_m) \cap (PBG = T_s)$
2. $F_s = 2\ F_m$: Critical sampling
   Recovery:
   $(f_s = 2f_m) \cap (f_c = f_m) \cap (PBG = T_s)$
3. $F_s < 2\ F_m$: Undersampling-> Recovery is not possible

Lowpass Sampling Theorem

"Lowpass signals can be sampled and recovered from their samples only if".

$F_s > 2F_m$

Proper LPF

&

f_s: Sampling frequency and sampling rate

f_m: Maximum frequency component present in m(t)

NYQUIST RATE

Frequency = $2f_m$

Baseband:

Nyquist rate is minimum in sampling

Frequency $f_{NY} = 2f_{max}$ f_{max} *BASEBEND*

Bandpass

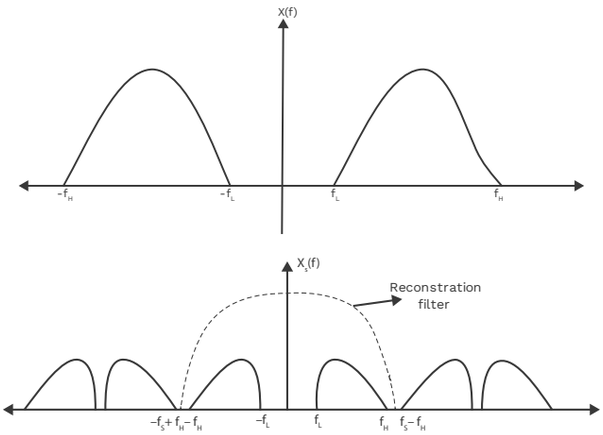
It is not necessarily the minimum sampling rate.

$f_{NY} = 2f_{max}$ $f_{max} \rightarrow$ *Bandpass*

Nyquist interval $\longrightarrow$ $T_{NY} = \frac{1}{f_{NY}}$

Baseband Sampling

$X(f) \xleftrightarrow{Sampling} X_s(f)$



⇒ Any baseband signal can be reconstructed from its samples by passing through low pass filter.

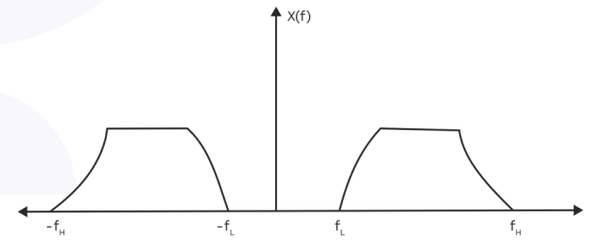
From above figure:

For proper reconstruction:

$f_s - f_H \geq f_H$

$f_s \geq (2f_H)$ where $2f_H \longrightarrow$ Nyquist rate

Bandpass Sampling



⇒ Any Bandpass signal can be reconstructed from its samples by passing it through Bandpass filter.

Sampling frequency (f):

Case 1:

$\frac{f_H}{B}$ = Integer

$f_s \geq 2B$

(B=Band width = $f_H - f_L$)

Case 2:

If $\frac{f_H}{B}$ = Integer = K

$K = \left[\frac{f_H}{B}\right]^{\text{greater interger function}}$ = Integer part

$$\frac{f_H}{M} \le f_s \le \frac{2f}{M-1} \qquad \boxed{M = 1, 2, ...K}$$

Case 3:

If $f_H >> B$

$$\boxed{f_s \ge 2B}$$

Example:

A Bandpass signal with f_H = 20KHz and f_L = 10KHz is sampled. Calculate the sampling rate to avoid Aliasing.

Solution:

$B = f_H - f_L$

B = (20-10) KHz = 10KHz

$$\frac{f_H}{B} = \frac{20}{10} = 2$$

$$\frac{f_H}{B} = \text{int}\,eger$$

Hence,

$f_s \ge 2B$

$f_s \ge 2 \times 10$ KHz

$$\boxed{f_s \ge 20KHz}$$

Sampling rate, $f_s \ge 20$ KHz

Key points to avoid mistake:

Do not blindly use $f_s = 2f_{max}$, for Nyquist sampling frequency. First check sampling, if it is bandpass then use above formula for frequency.

- A Bandpass signal can be directly transmitted through a wired channel, so a wired channel is also called as Base band channel or low-pass channel.
- Baseband signal can be directly transmitted through free space, so free space is also called as bandpass channel.

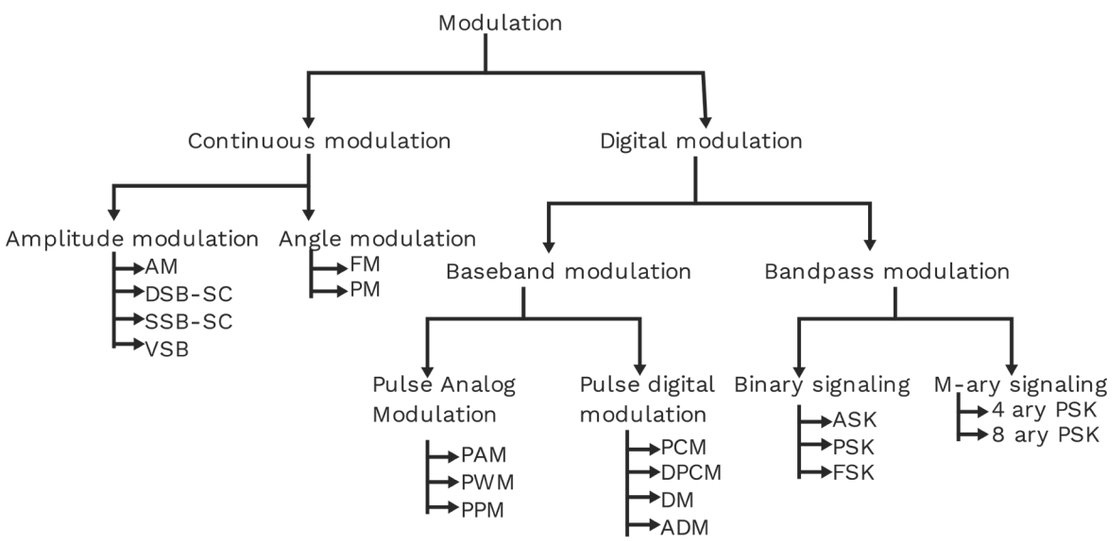
Pulse Analog Modulation

As we learned previously, in analog modulation systems, different parameters of a sinusoidal carrier are varied with respect to the instantaneous value of the modulating signal. While in pulse analog modulation, the carrier consists of a pulse train. Some parameters of which are varied according to the instantaneous value of the modulating signal.

Pulse Amplitude Modulation (PAM)

The amplitude of the carrier pulse train is varied with respect to the modulating signal. i.e., the amplitude of the pulse depends on the value of m(t) during the time of pulse.

Types of modulation



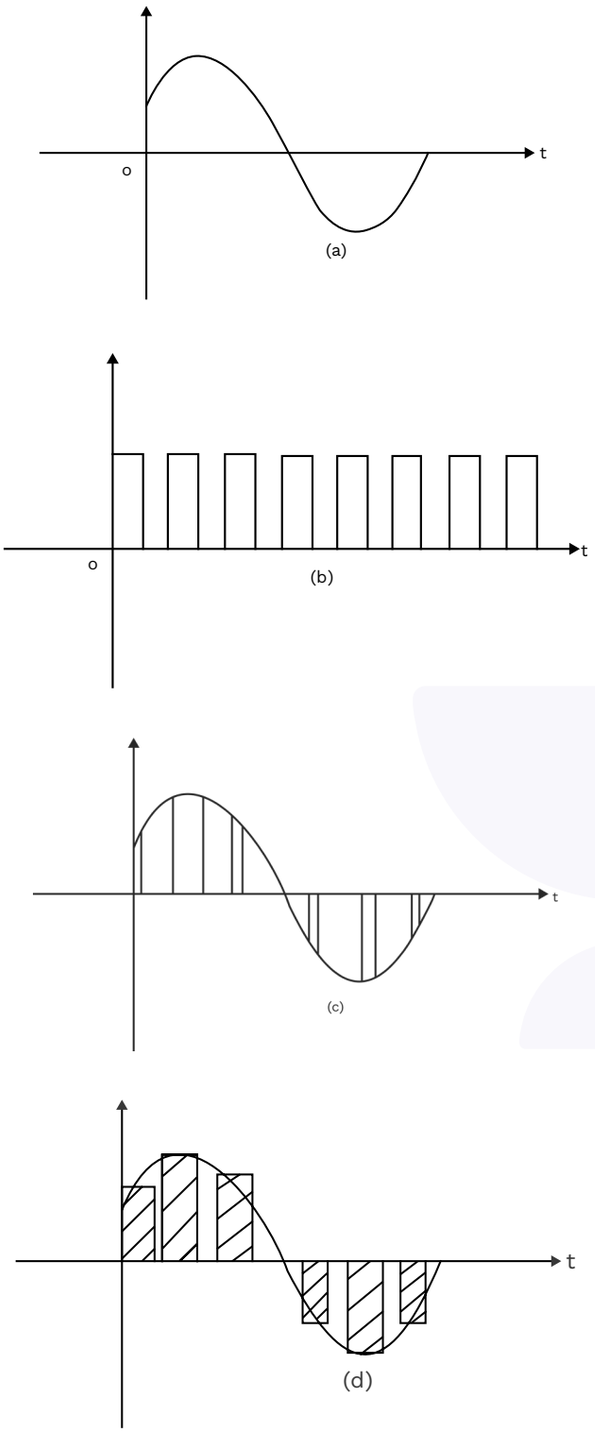


Fig. 5.3 (A) Modulating Signal (B) Carrier Pulse Train (C) Natural PAM (D) Flat-top PAM

In natural PAM, a signal is sampled at the Nyquist rate and reconstructed by passing it through a Low Pass Filter (LPF) with a precise cutoff frequency. Though the PAM signal is passed through an LPF, it cannot recover the signal without distortion. Hence, to avoid this, flat-top sampling is done, as shown in the above figure.

Pulse Width Modulation (PWM)

Pulse Width Modulation (PWM) is an analog modulating scheme in which the width or duration of the pulse carrier varies with respect to the instantaneous amplitude of the message signal, i.e., the width of the pulse varies in this method, but amplitude remains constant.

Pulse Position Modulation (PPM)

In this, amplitude and width of the pulse are constant, while the position of each pulse with reference to the position of a reference pulse varies according to the instantaneous sampled value of the message signal.

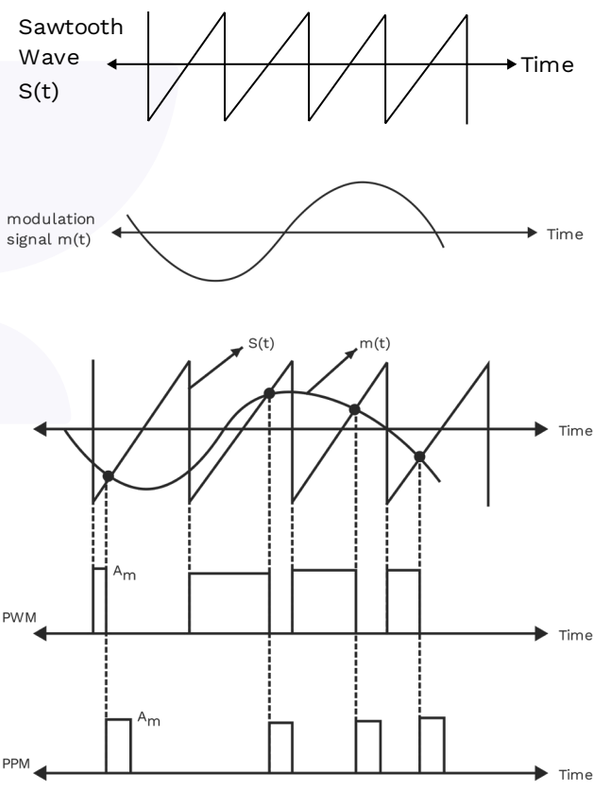


Fig. 5.4 Generation of PWM and PPM

Quantization Process

In a continuous signal, for a fixed range of the signal we find an infinite number of amplitude levels. It is not necessary, in fact, to transmit the exact amplitudes of the samples. Any human sense (an ear or eye) as the

ultimate receiver can detect only finite intensity differences. Due to this, the original continuous signal can be approximated by a signal constructed of discrete amplitude selected on a minimum error basis from the available set.

Amplitude quantization is defined as the process of transforming a sample amplitude $x(nT_s)$ of a message signal x(t) at time $t = nT_s$ into a discrete amplitude, taken from a finite set of possible amplitudes.

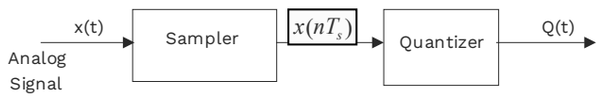


Fig. 5.5 Block Diagram for Quantization

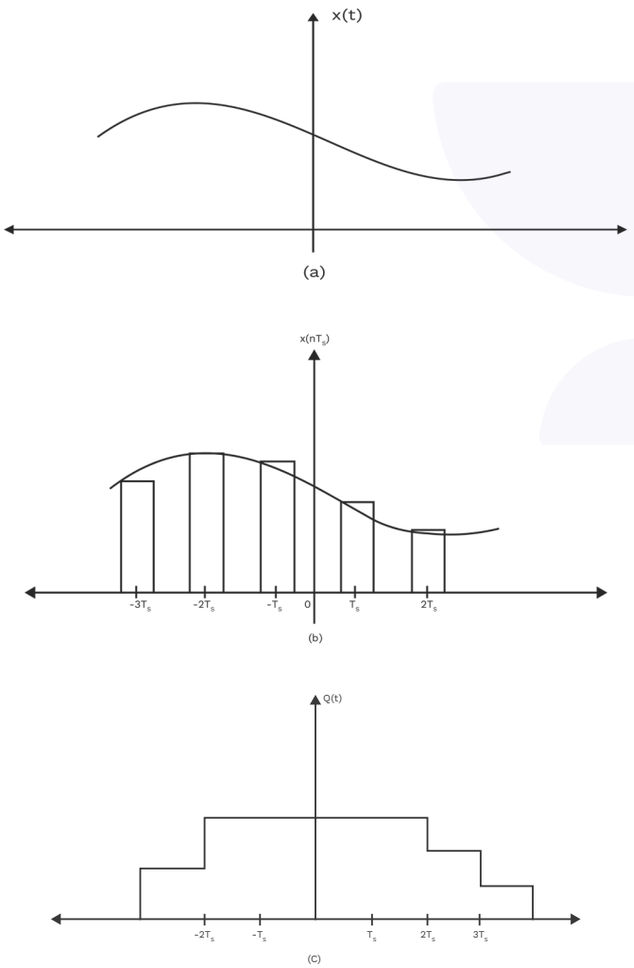


Fig. 5.6 (A) Analog Signal (B) Sampled Signal (C) Quantized Signal

Quantization is necessary process for pulse code modulation. The signal amplitude $\hat{a}$ is specified by index i, if it is lies inside the partition cell.

$Q_i = \{a_i < a < a_{i+1}\}$ i = 1, 2, 3 ... L

(L = Number of levels)

The discrete amplitude a_i represented the decision levels or decision thresholds. At the quantizer output, the index i is transformed into an amplitude $\hat{x}_k$ where, k = 1,2,3...L are called representation levels or reconstruction levels and the spacing between two adjacent representation levels is known as step size.

Thus, the quantizer output $\hat{x}$ equals $\hat{x}_i$ if the input signal sample belongs to the interval Q_i. The mapping shown in below Figure is the quantizer characteristics, which is a staircase function by definition.

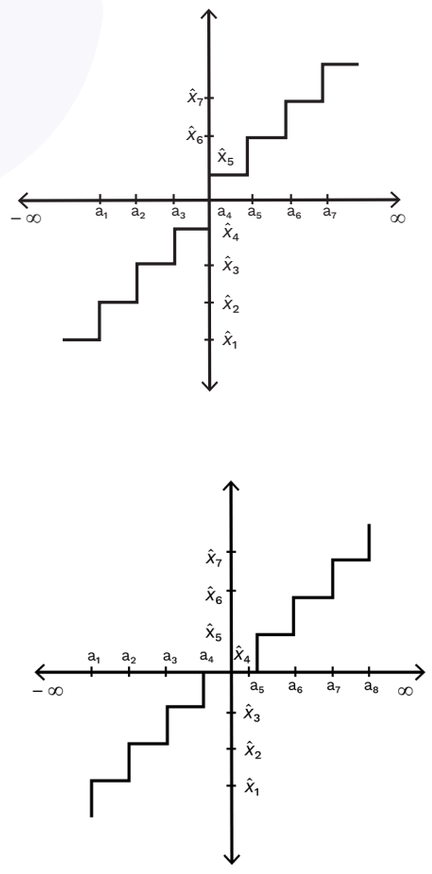


Fig. 5.7 (A) Midrise Quantization (B) Midtread Quantization

Step size:

Spacing between two adjacent decision boundary

$\Delta_i = (a_i - a_{i-1})$

where

$\Delta_i = step - size$

a_i is decision boundary

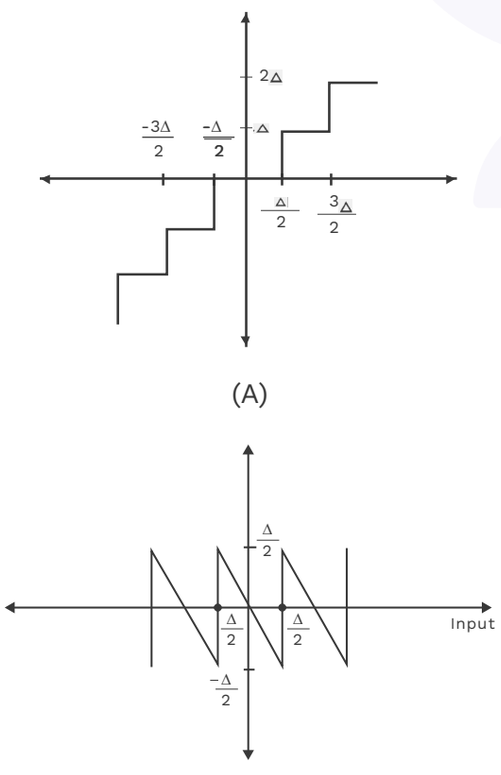
If step size of every quantization region is equal, then it is known as uniform quantization

$$\Delta = \frac{Dynamic\ range}{L}$$

$$\Delta = \frac{V_{p-p}}{L}$$

Quantization Noise or Quantization Error

$Q_{error} = Q_{noise} = (output - input)$



(A)

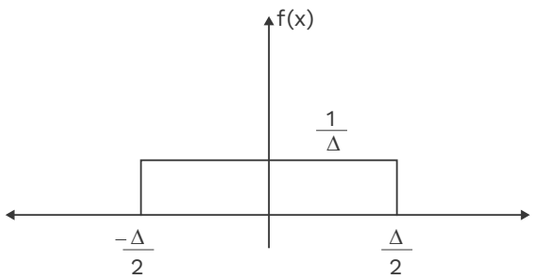
(B)

Fig. 5.8 (A) Quantizer (B) Quantization Noise

As $Q_{noise} = (output - input)$

And if Δ is step size, the Q_{noise} or Q_{error} will have value always between $-\frac{\Delta}{2}$ to $\frac{\Delta}{2}$ with equal probability.

Now, PDF of Q_{noise}



- Mean of quantization noise = $E[Q_{noise}] = 0$
- Quantization noise power = $E[Q^2_{noise}]$

$= \frac{(b-a)^2}{12}$ (For uniform distribution)

Here $b = \frac{\Delta}{2}, a = -\frac{\Delta}{2}$

$$E[Q^2_{noise}] = \frac{\left(\frac{\Delta}{2} + \frac{\Delta}{2}\right)^2}{12} = \frac{\Delta^2}{12}$$

Quantization noise power $= \frac{\Delta^2}{12}$

Example:

A random variable X with uniform density in the interval 0 to 2 is quantized as follows:

If $0 \le X \le 0.9$, $x_q = 0$

If $0.9 < X \le 2$, $x_q = 1.5$

Where x_q is the quantized value of X. Calculate the root mean square value of quantization noise.

Solution:

Root mean square value

$= \sqrt{mean\ square\ value}$

Mean square value of quantization noise

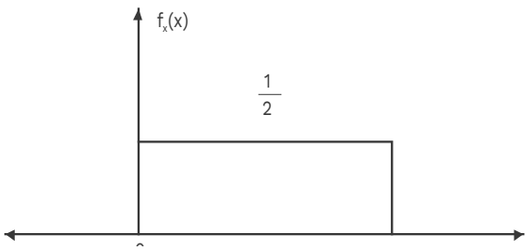
$= E[Q^2_{noise}]$

$= \int_{-\infty}^{\infty} Q^2_{noise} f_x(x)dx$

$= Q_{noise} = output - input$

$= Q_{noise} = \begin{cases} (0-x) & For\ 0 \le x \le 0.9 \\ (1.5-x) & For\ 0.9 < x \le 2 \end{cases}$

$= E\left[Q^2_{noise}\right] = \int Q^2_{noise} f_x(x)dx$



$$E\left[Q^2_{noise}\right] = \int_0^{0.9} X^2 \times \frac{1}{2}dx + \int_{0.9}^{2} (1.5-x)^2 \times \frac{1}{2}dx$$

$$= \frac{1}{2}\left[\frac{x^3}{3}\right]_0^{0.9} + \frac{1}{2}\int_{0.9}^{2} (2.25 + x^2 - 3x)dx$$

$$= \frac{1}{2}\left[\frac{(0.9)^3}{3} + \left[2.25x + \frac{x^3}{3} - \frac{3x^2}{2}\right]_{0.9}^{2}\right]$$

$$= \frac{1}{2}\left[\begin{array}{l} 0.243 + 2.25 \times 2 + \frac{(2)^3}{3} - \frac{3 \times 4}{2} - \\ 2.25 \times 0.9 - \frac{(0.9)^3}{3} + \frac{3 \times (0.9)^2}{2} \end{array}\right]$$

$$= \frac{1}{2}\left[2.475 + 2.667 - 6 + 1.215\right]$$

Mean square value = 0.1785

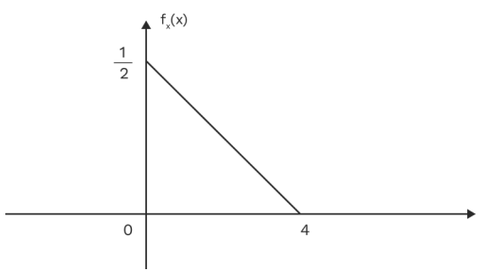
Root mean square value of quantization noise $= \sqrt{mean\ square\ value}$

$= \sqrt{0.1785}$

Root means square value = 0.422

Example:

A random variable X is distributed as shown in given PDF $f_x(x)$



Is quantized as follows

If X < 2 $X_q = 1$

If X > 2 $X_q = 3$

Where x_q is the quantized value of X. Calculate the quantization noise power.

Solution:

Quantization noise power

$= E\left[Q^2_{noise}\right]$

$= \int_{-\infty}^{\infty} Q^2_{noise} f_x(x)dx$

From the given PDF:

$$f_x(x) = -\frac{1}{8}x + \frac{1}{2}$$

$$Q_{noise} = \begin{cases} (1-x) & X < 2 \\ (3-x) & X > 2 \end{cases}$$

Quantization noise power

$$= \int_0^2 (1-x)^2\left[-\frac{1}{8}x + \frac{1}{2}\right]dx + \int_2^4 (3-x)^2\left[-\frac{1}{8}x + \frac{1}{2}\right]dx$$

$$= \int_0^2 (1 + x^2 - 2x)\left[-\frac{1}{8}x + \frac{1}{2}\right]dx + \int_2^4 (9 + x^2 - 6x)\left[-\frac{1}{8}x + \frac{1}{2}\right]dx$$

$$= \int_0^2 \left(\frac{-9}{8}x - \frac{1}{8}x^3 + \frac{3}{4}x^2 + \frac{1}{2}\right)dx + \int_2^4 \left(\frac{-1}{8}x^3 + \frac{5}{4}x^2 - \frac{33}{8}x + \frac{9}{2}\right)dx$$

$$= \left[\frac{-9}{16}x^2 - \frac{x^4}{32} + \frac{1}{4}x^3 + \frac{1}{2}x\right]_0^2 + \left[\frac{-x^4}{32} + \frac{5}{12}x^3 - \frac{33}{16}x^2 + \frac{9}{2}x\right]_2^4$$

$$= \left[\frac{-9}{4} - \frac{1}{2} + 2 + 1\right] + \left[\begin{array}{l} -8 + \frac{5}{12} \times 64 - 33 + 18 + \\ \frac{1}{2} - \frac{40}{12} + \frac{33}{4} - 9 \end{array}\right]$$

= 0.25 + 0.0833

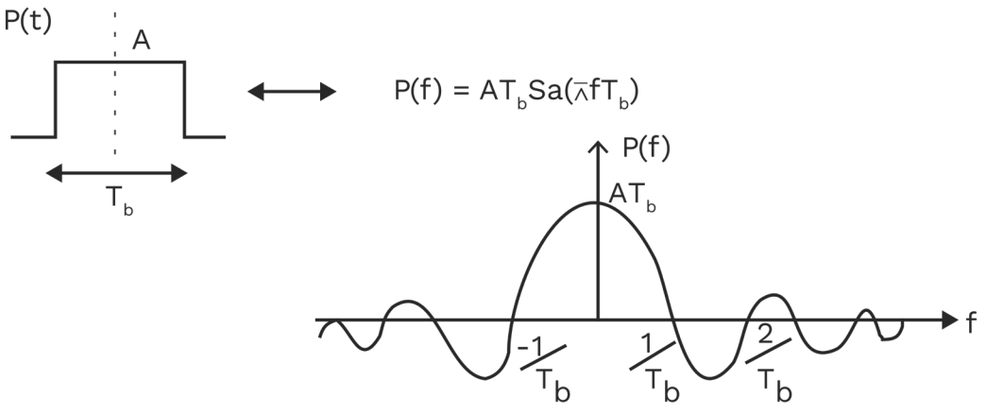
= 0.33

Quantization noise power = 0.33

Pulse Transmission

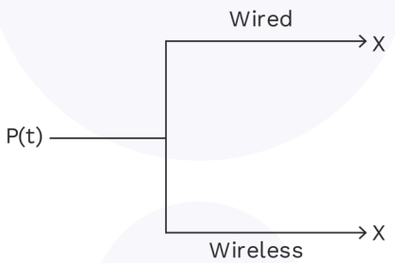
1. Rectangular pulse
2. Sinc pulse

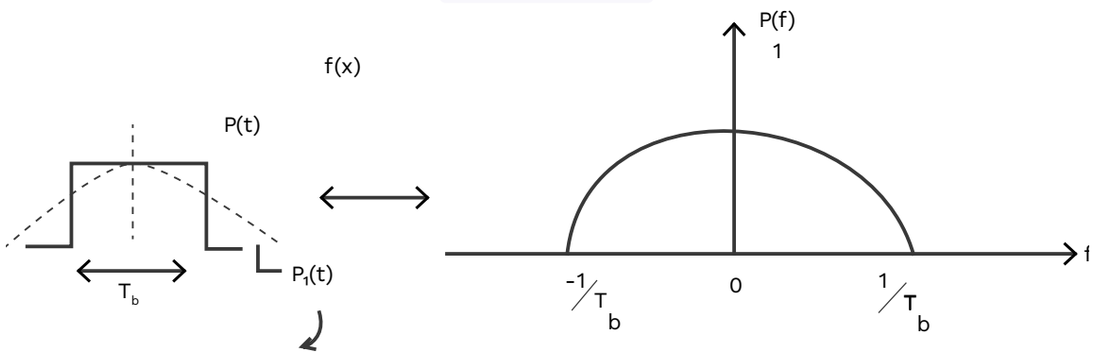
Rectangular pulse:



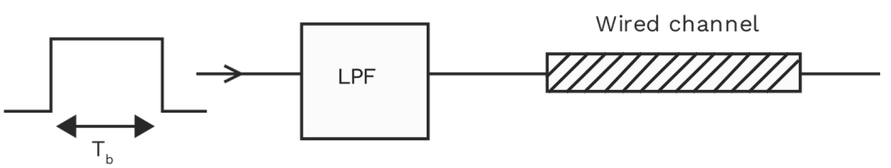
p(t) : Time Limited → Band unlimited

: Baseband





After Bandlimiting



Bandwidth of wired channel:

$$Bw \geq \frac{1}{T_b}$$

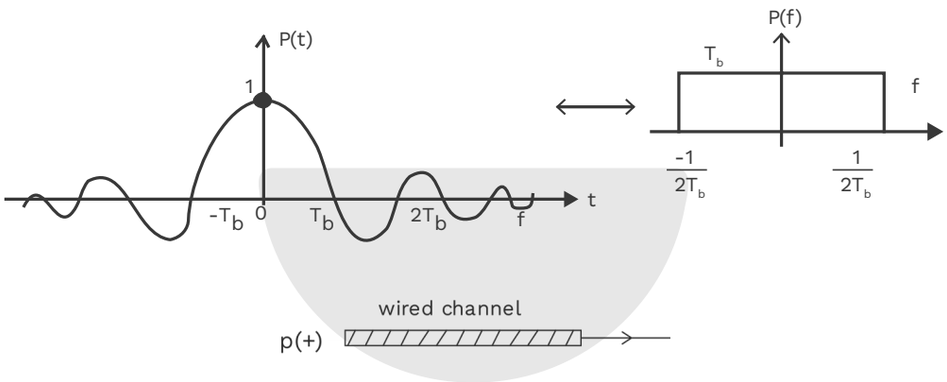
$$Bw \geq \frac{1}{2T_b} \rightarrow \text{Bit Rate}$$

Sinc pulse:

p(t) = having zero-crossovers of T_b p(t) = sinc (t/Tb)

$$p(t) \xrightarrow{\text{wired channel}}$$

$$Bw \geq \frac{1}{2T_b}$$



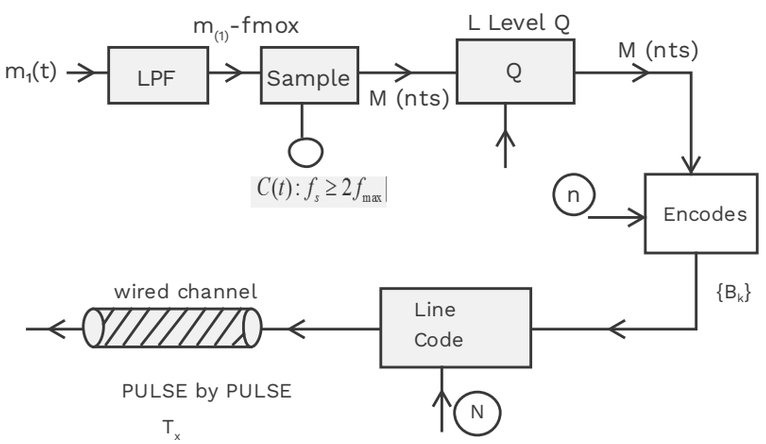
$$\boxed{(Bw) \geq \frac{R_b}{2}}$$

Note:

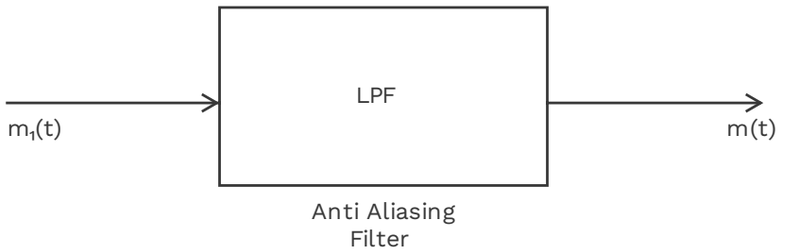
1. To transmit a Rect pulse of duration T_o via wired channel → $\boxed{Channel\ Bw \geq R_b}$
2. To transmit a sinc pulse of zero-crossover at KT_b via wired channel → $\boxed{Channel\ Bw \geq \frac{R_b}{2}}$

Analog to Digital Converter (ADC)

Let m(t) is an analog signal

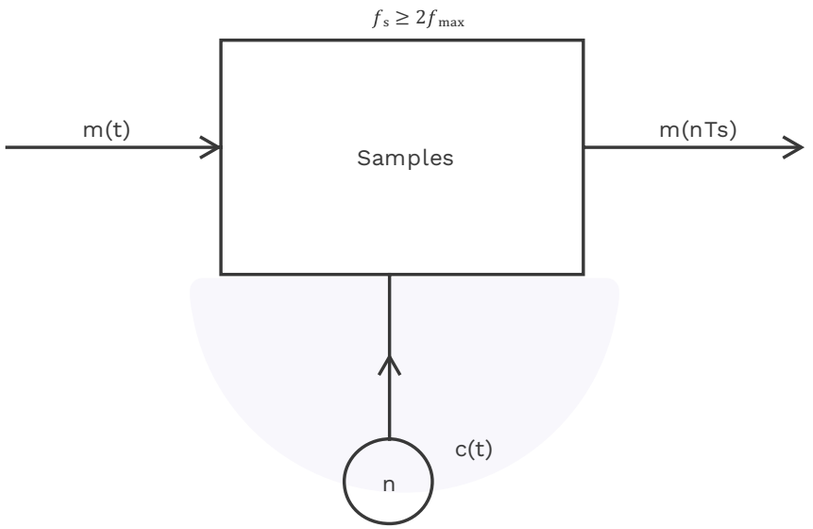


1.

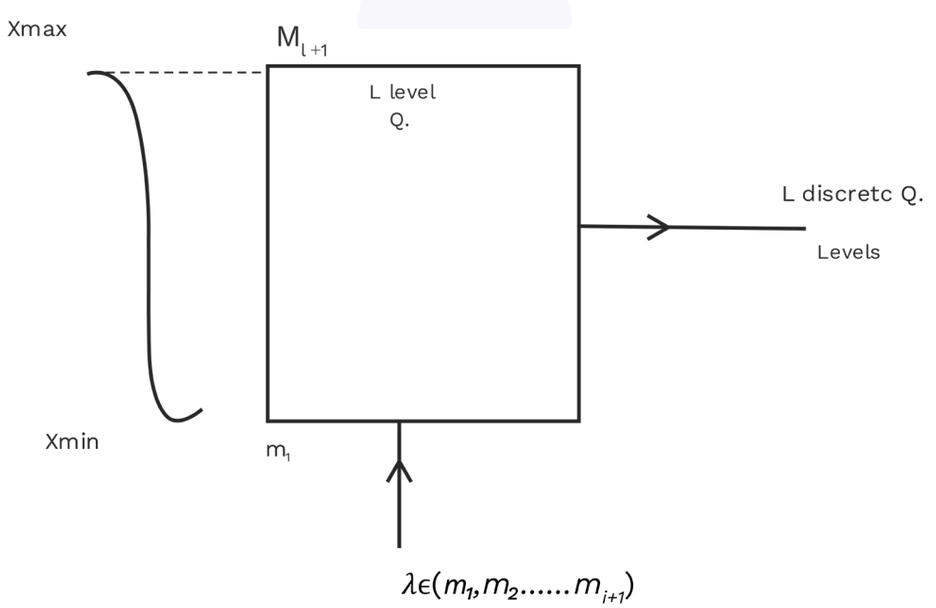


Anti Aliasing Filter

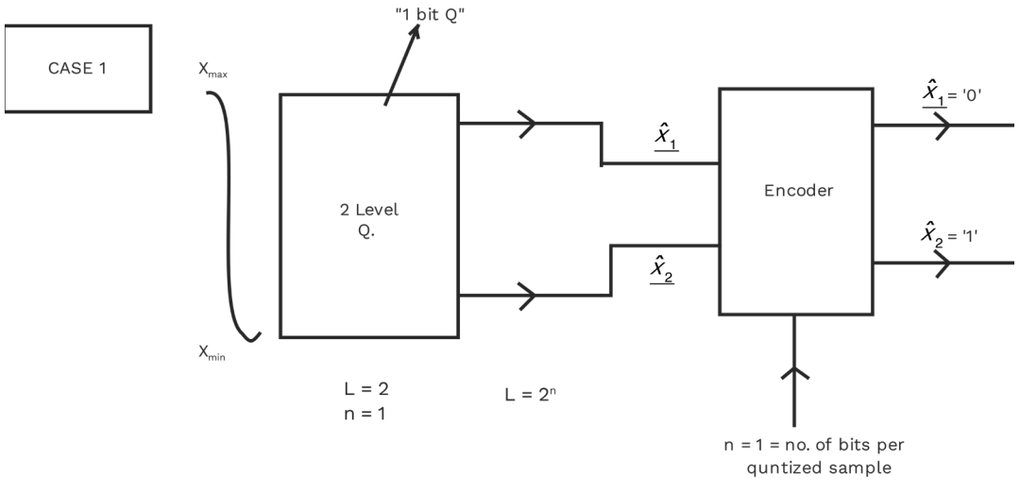
2.

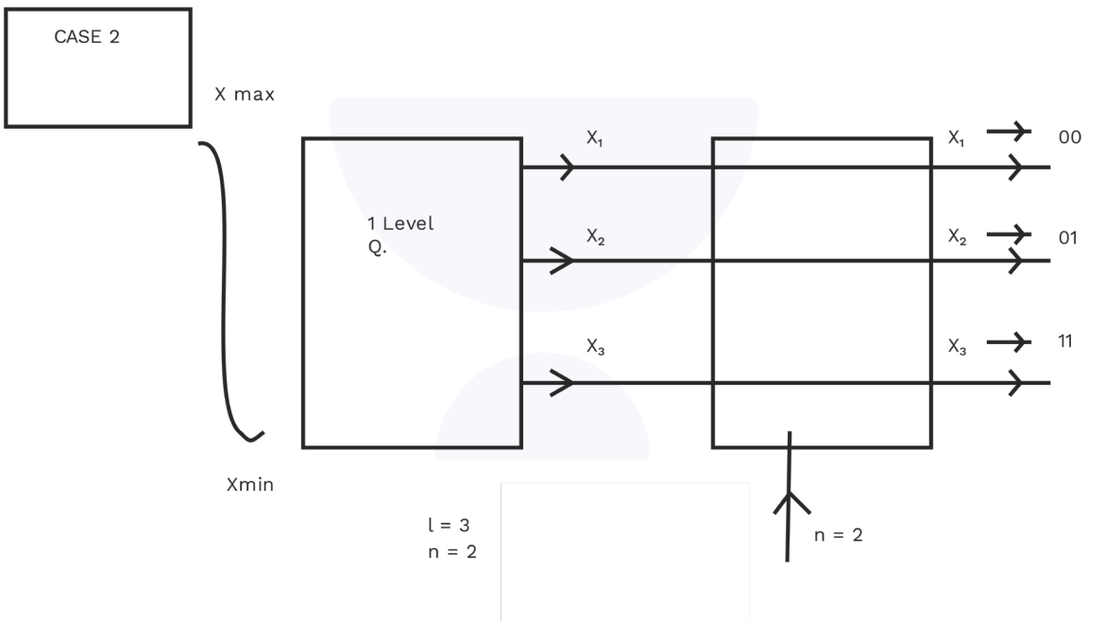


3.



4. Encoder



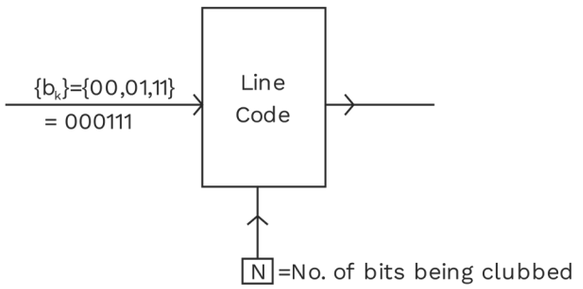


a) A Q is also defined as n bit Q, which mean $L = 2^n$.

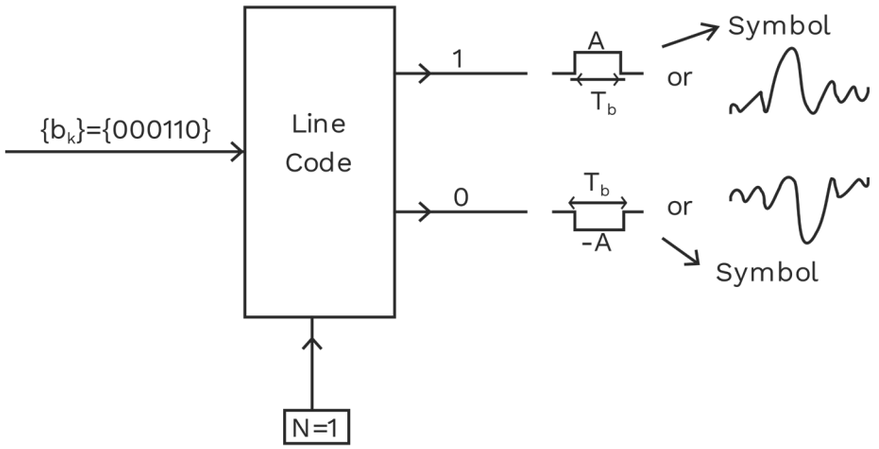
b) n = No. of bits assigned to each quantized sample

$\boxed{L \leq 2^n}$ $\boxed{L_{max} = 2^n}$

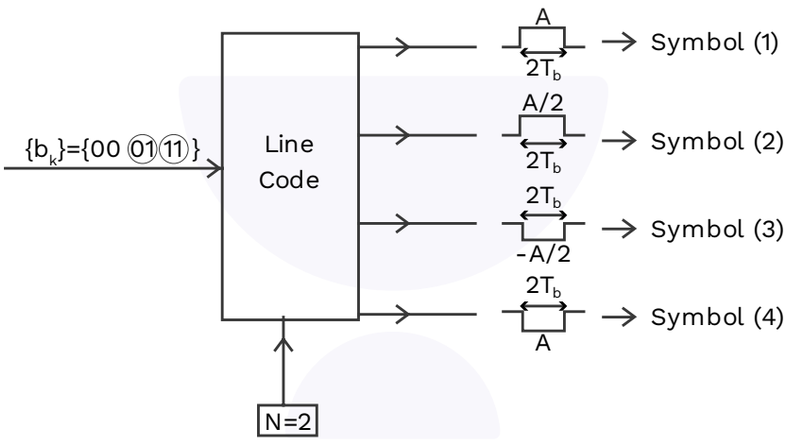
Line coder:



CASE 1: 2 Ary scheme

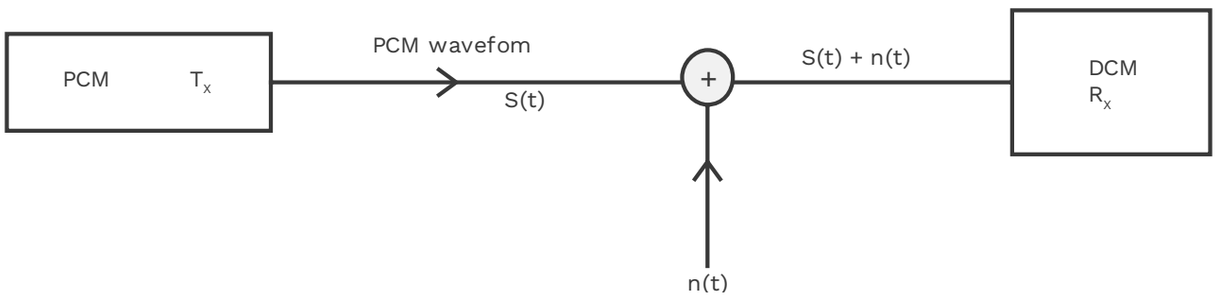


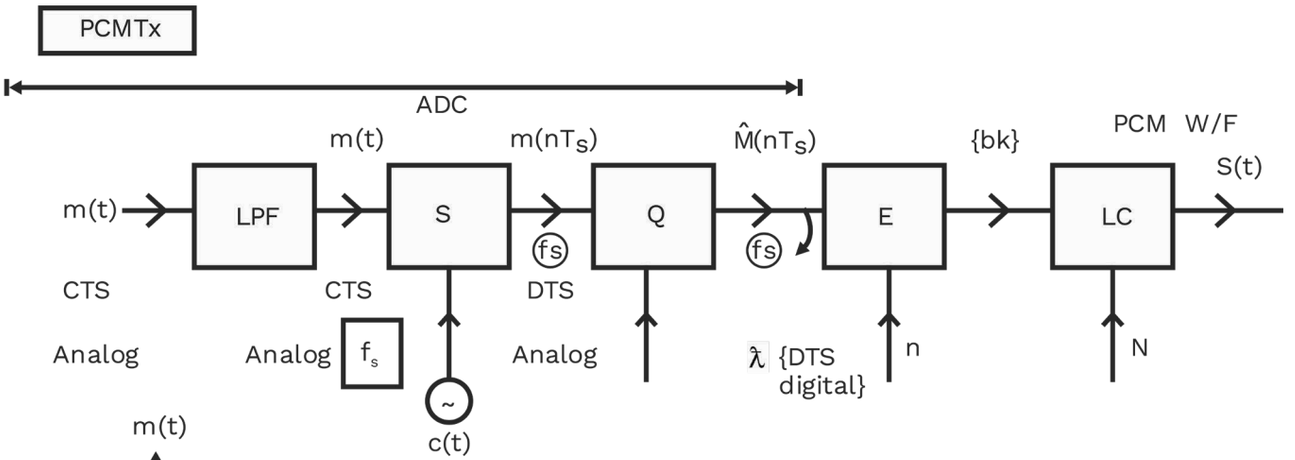
CASE 2: 4 Ary scheme

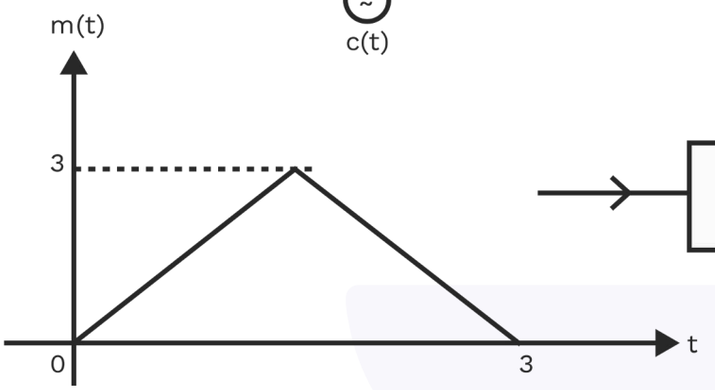


Pulse Code Modulation

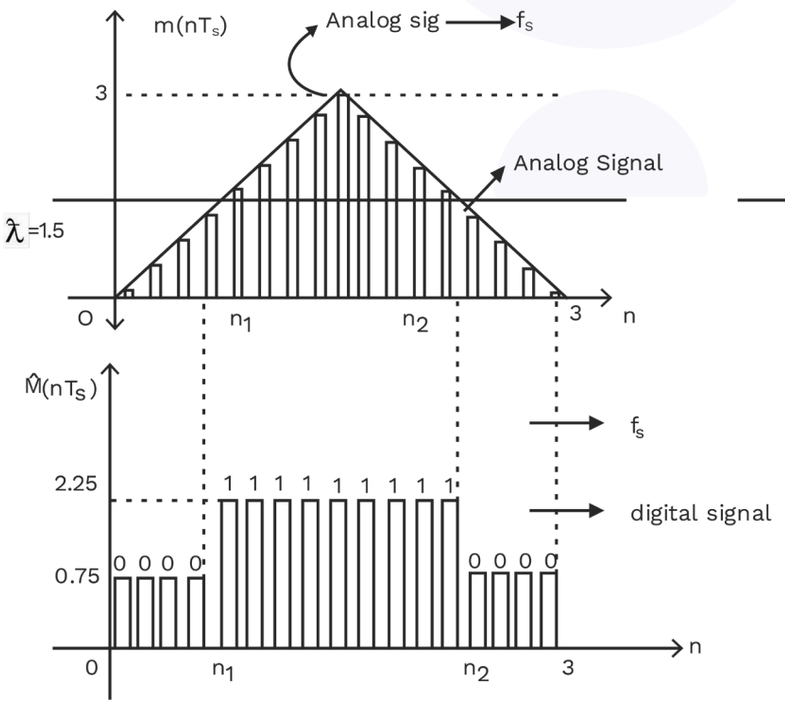
In pulse code modulation, the message signal is represented by a sequence of coded pulses. It is a digital technique that involves sampling an analog signal at regular intervals and coding the measured amplitude into a series of binary values which are transmitted by modulation of a pulse. In this process we use uniform quantizer with step size Δ.

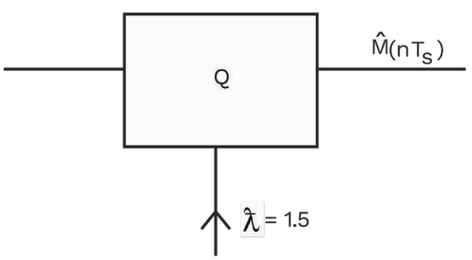


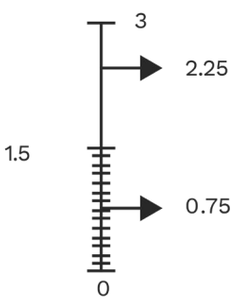




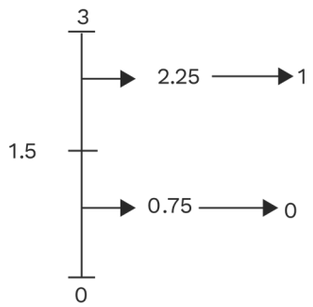




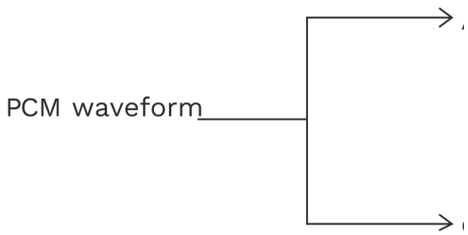




Note: Sampling interval of sampled signal and quantized signal remains same.



Encodes



Pulse by pulse Tx: Each pulse should be Txed by wired channel.

Ary scheme: (B.W. of channel) $\geq R_b \rightarrow$ Rect

Or

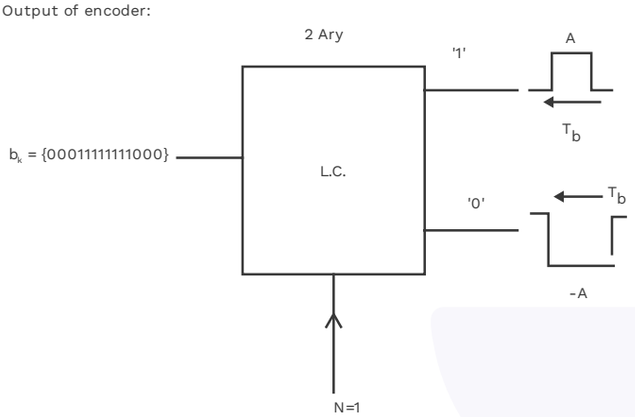
$\geq \frac{R_b}{2} \rightarrow$ Sinc

M Ary scheme: (B.W. of channel)

$\geq \frac{1}{T_s} = \frac{R_b}{N} \rightarrow Qiect$

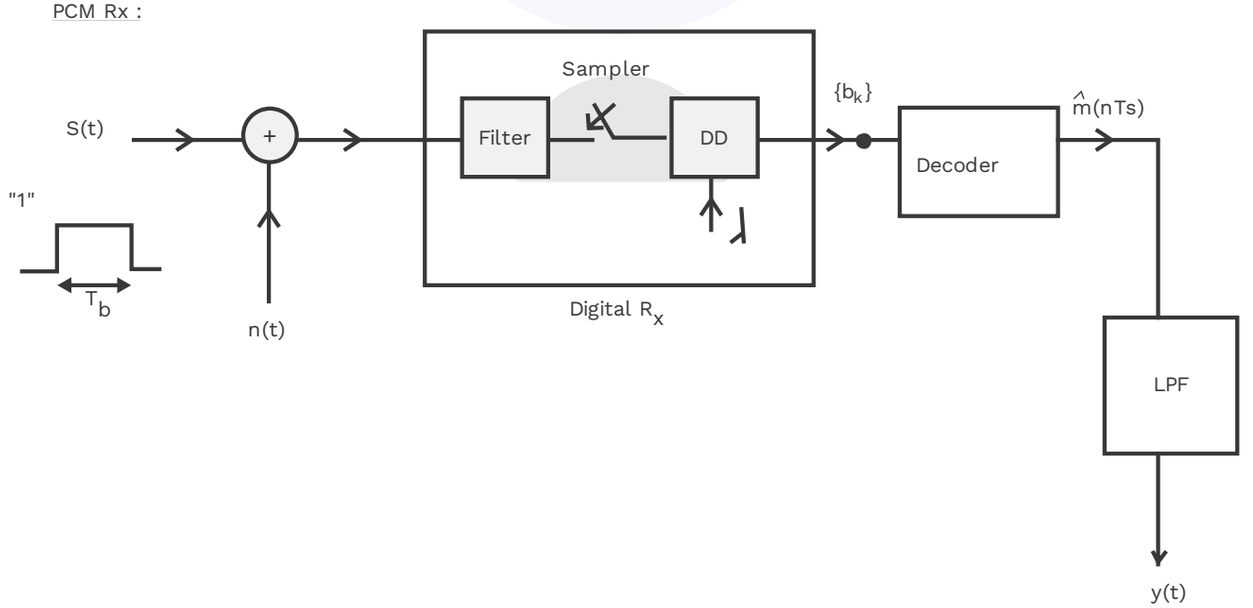
$\geq \frac{R_b}{2N} \rightarrow$ Sinc

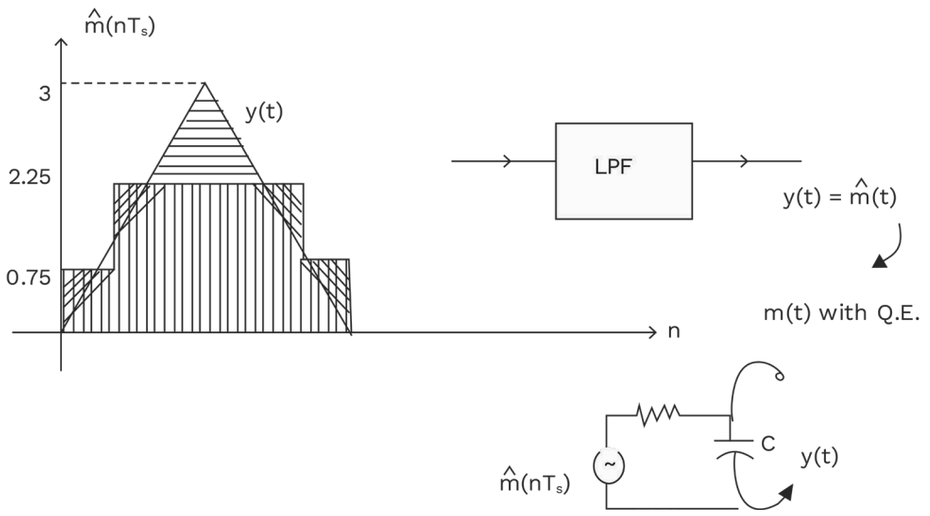
Output of encoder:



PCM Rx:

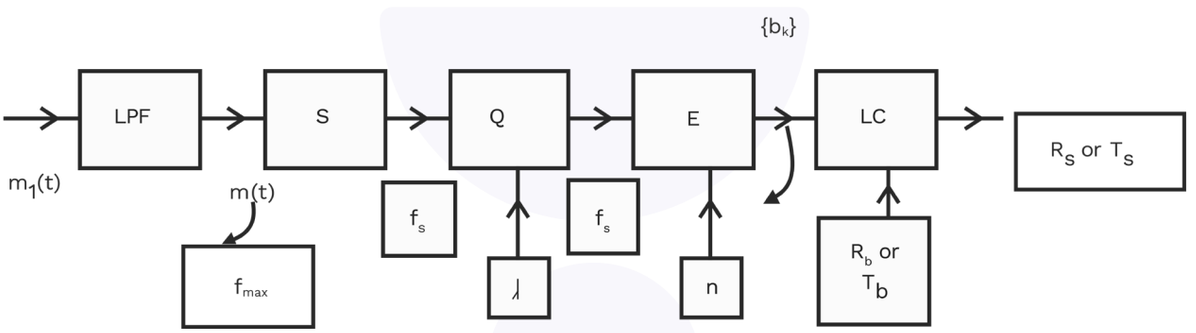
PCM Rx :





Note:

1. Without Quantizer, encoding becomes impractical.
2. Quantizer is a non-invertible system or device.



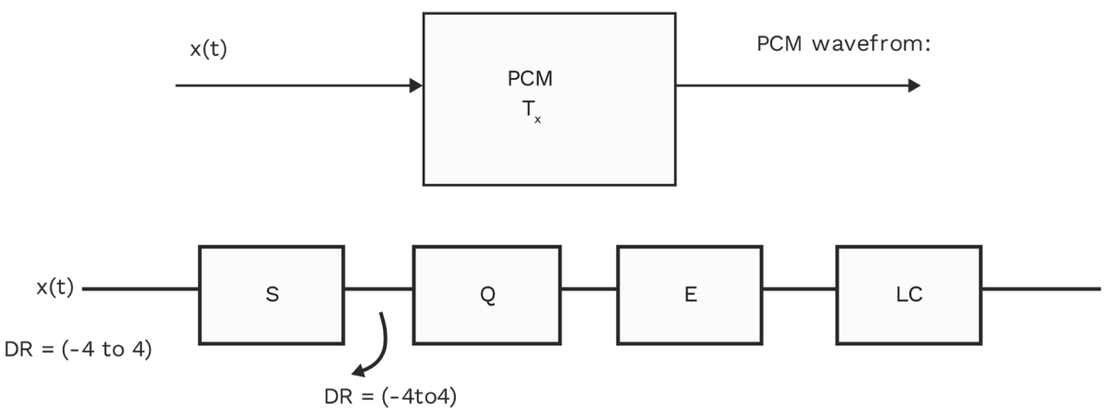
n = No. of bits/sample

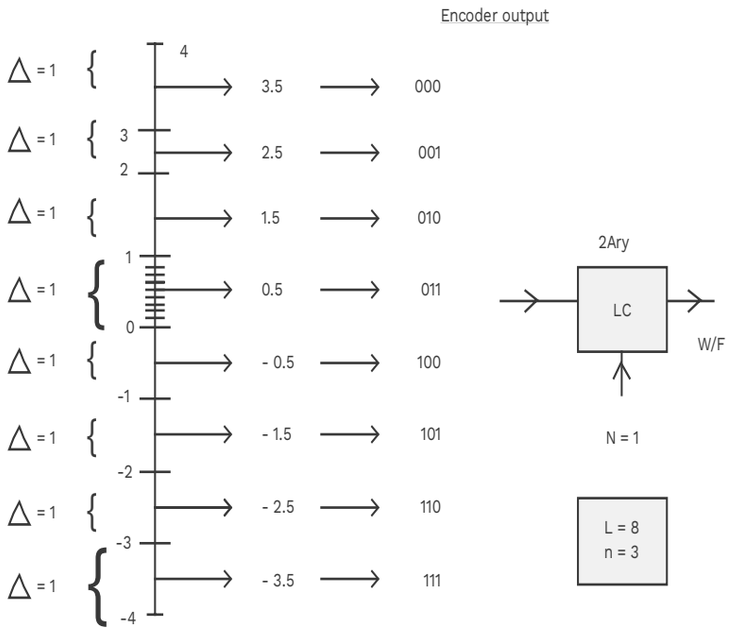
f_s = Samples/sec

$R_b = nf_s$ = Bits/sec

Example: x(t) = 4 sint

$\Delta = 1v$





Sv = 0.3 Qv = 0.5 Q.E.= Sv- Qv = -0.2

Sv = 0 Qv = $\begin{matrix}0.5\\0.5\end{matrix}$ Q.E. = ± 0.5 = ± $\Delta/2$

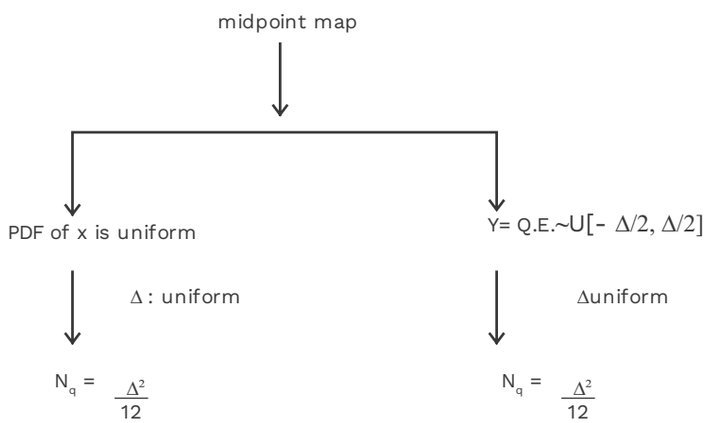
$\boxed{\max Q.E. = \pm \Delta/2}$ → Only when mid-point map is considered.

Let midpoint map is allowed and R.V. $Y = QE = \begin{cases} defined & -\Delta/2 \le y \le \Delta/2 \\ 0 & else \end{cases}$

$$Y \sim U\left[\frac{-\Delta}{2}, \frac{\Delta}{2}\right]$$

$$E\left[x^2_{QE}\right] = \frac{\left(\Delta/2\right)^2}{3} = \frac{0^2}{12}$$

Note :



Formula of PCM

Let : Uniform Quantization mid point map

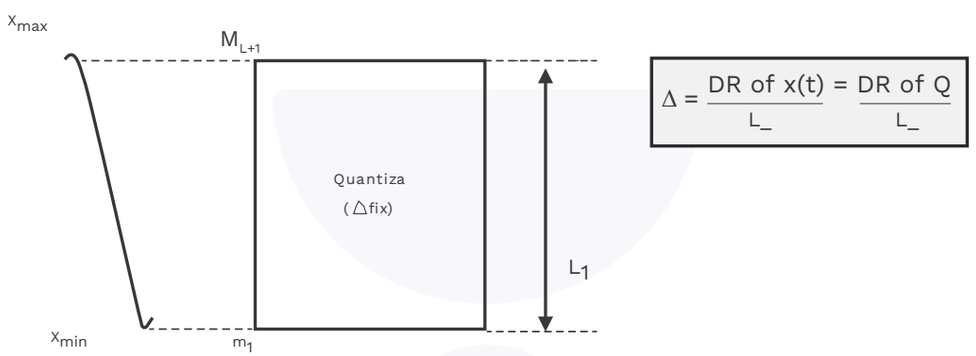
$X_{QE} = Y : U\left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right]$

$$N_q = E\left[x^2_{QE}\right] = \frac{\Delta^2}{12}$$

1. Let Analog signal X(t) utilize all quantization levels

   (Or)

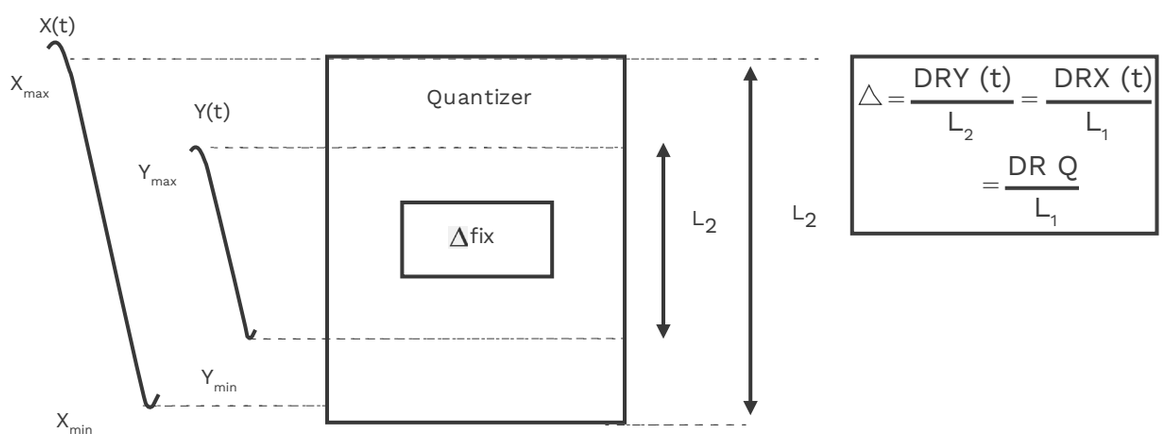
   Signal fully loads the quantize



2. Let Y(t) does not utilize all quantization levels.

   (Or)

   Y(t) does not fully load quantizer.



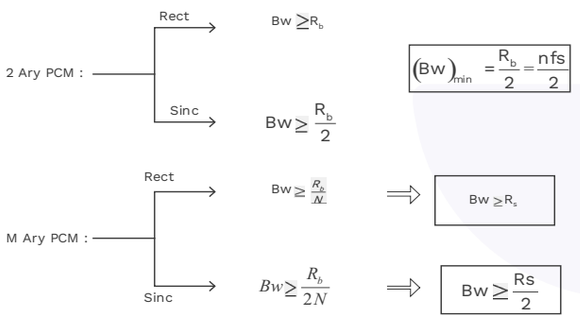
3. $\Delta = \frac{\text{DR of Q}}{L_1}$

4. $QE_{max} = \pm\frac{\Delta}{2}$

5. $L \le 2^n \rightarrow L_{max} = 2^n$
6. $P_s = \overline{x^2(t)}$ or $P_s = E\left[x^2(t)\right]$
7. $P_{QE} = E\,[x^2_{QE}]$
8. $SQNR = \dfrac{P_s}{P_{QE}}$ $\quad SQNR\,(dB) = 10 \log_{10} \dfrac{P_s}{P_{QE}}$
9. Bit Rate = $R_b = nf_s$
10. Bit Interval = $T_b = \dfrac{1}{R_b}$
11. Bard rate or Symbol rate

$$\boxed{\begin{matrix} T_s = NT_b \\ R_s = R_b / N \end{matrix}} \rightarrow \text{MAry scheme}$$

12. Transmission Bw



Calculation of signal power to quantization noise power ratio:

Let $P_{QE} = E[x^2_{qE}] = \dfrac{\Delta^2}{12}$

Case 1: $m(t) = A_m \sin \omega_m (t)$

$\rightarrow P_s = \dfrac{A_{m^2}}{2}$

$\rightarrow P_{QE} = \dfrac{\Delta^2}{12}$ $\quad \Delta = \dfrac{DR}{L} = \dfrac{2A_m}{L} \rightarrow \Delta^2 = \dfrac{4A_m{}^2}{L^2}$

$P_{QE} = \dfrac{A_{m^2}}{3L^2}$

$$\boxed{SQNR = \frac{P_s}{P_{QE}} = \frac{3L^2}{2}}$$

$SQNR\,(dB) = 10 \log .1.5 + 20 \log_{10} L$

$$\boxed{SQNR\,(dB) = (1.76) + 20 \log_{10} L}$$

maximum SQNR
Or
Signal utilizes all Q Levels
Or
Signal Fully Loads the Q

$SQNR = \dfrac{3L^2}{2}$

$(SQNR) \le \dfrac{3}{2}(4)^n$

$L \le 2^n$

$L_{max} = 2^n$

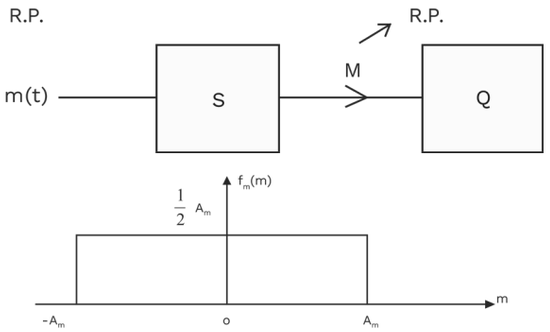
$(SQNR)_{max} = \dfrac{3}{2}(4)^n$

$$\boxed{(SQNR)_{max} dB = (1.76 + 6n) dB}$$

SUMMARY

1. $\boxed{SQNR = \dfrac{3}{2}L^2}$
2. $\boxed{SQNR \le \dfrac{3}{2}(4)^n}$
3. $\boxed{(SQNR)_{max} = \dfrac{3}{2}(4)^n}$

Case 2: Let m(t) is uniformly distributed between $-A_m$ to A_m.



$SQNR = \dfrac{P_s}{P_{QE}}$

$P_s = E[m^2(t)] = E[M^2] = \dfrac{A_m{}^2}{3}$

$P_{qE} = \frac{\Delta^2}{12} = \frac{Am^2}{3L^2}$ Since $\Delta = \frac{2A_m}{L}$

$\boxed{SQNR=L^2}$

SUMMARY

SQNR = L^2 $L \le 2^n$	1) SQNR(dB) = $20\log_{10} L$
SQNR ≤ 4^n	2) SQNR(dB) ≤ 6n
$(SQNR)_{max} = 4^n$	3) $(SQNR)_{max}$ = 6n

Some Important Observations

1. If n is given in problem
   $L=2^n$
   $SQNR = (SQNR)_{max}$
2. If L is mentioned in problem
   $L \le 2^n \rightarrow n_{min}$ = ?
3. If L is mentioned in problem
   L= Binary power = $2^n \rightarrow n$ = ?
   L≠ Binary power ≤ $2^n \rightarrow n_{min}$
4. $\boxed{\Delta = \frac{DR}{L}}$
5. $\boxed{SQNR = \frac{E(X^2)}{E(X^2_{qE})} = \frac{P_s}{P_{QE}}}$
6. If $P_{QE} = \Delta^2/12$



Example:

A signal having a bandwidth equal to 3.5 kHz is sampled, quantized and coded by a PCM system. The coded signal is then transmitted over a transmission channel that supports a transmission rate of 50×10^3 bits/second. Determine the maximum signal-to-noise ratio that can be obtained by this system. The input signal has a peak-to-peak value of 4 volts and an RMS value of 0.2 volt.

Solution:

$(SQNR)_{max} = \left(\frac{P_s}{P_{QE}}\right)_{max}$ f_m = 3.5 KHz.

$R_b = 50\times10^3$

DR = 4V

$\sqrt{m^2(t)} = 0.2V \rightarrow P_s = 0.04\,watt$

$$\boxed{P_{QE} = \frac{\Delta^2}{12} = \frac{DR^2}{12L^2} \qquad P_{QE} = \frac{16}{12L^2} = \frac{4}{3L^2}} \qquad \Delta = \frac{DR}{L}$$

$R_b = nf_s \ge 2nf_m$ $\boxed{f_s \ge 2f_m}$

$R_b \ge 2nf_m$ $n \in I$

$n \le \frac{50}{7} \rightarrow n \le 7.1 \rightarrow \boxed{n = 7} \rightarrow n_{max}$

$n_{max} = 7 \rightarrow L_{max} = 2^7 = 128$

$(P_{QE})_{min} = \frac{4}{3(L\,max)^2} = \frac{4}{3(128)^2}$

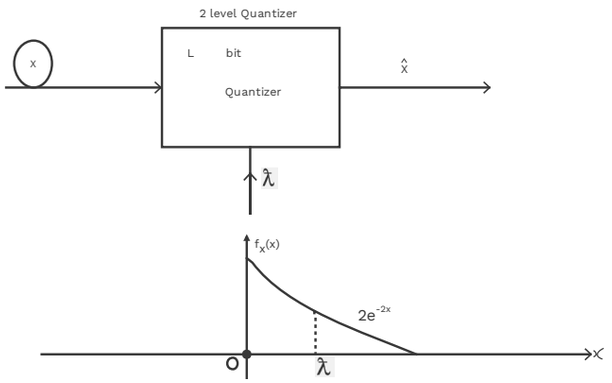
$\frac{P_s}{(P_{QE})_{min}} = (SQNR)_{max} = \frac{0.04}{\frac{4}{3(12\theta)^2}} = 491.52 = SQNR$

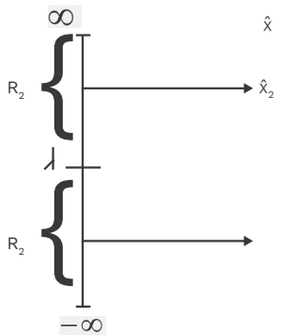
(SQNR) dB = $10\log_{10}$ (491.52) = 26dB

Example:

The input to a 1–bit quantizer is a random variable X with PDF $f_x(x) = 2e^{-2x}$ for $x \ge 0$ and $f_x(x) = 0$ for $x < 0$. For outputs to be of equal probability, the quantizer threshold should be_____.

Solution:





$$P\{\hat{x} = \hat{x}_1\} = P\{\hat{x} = \hat{x}_2\} = \frac{1}{2}$$

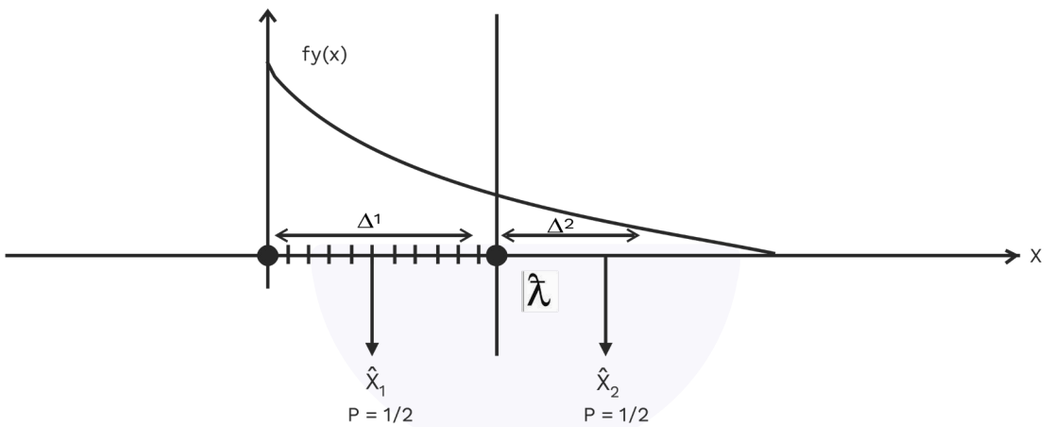
$$P\{\hat{x} = \hat{x}_1\} = P(-\infty < x < \lambda) = \int_{-\infty}^{\lambda} 2e^{-2x}u(x)dx$$

$$\int_0^{\lambda} 2e^{-2x}dx = \frac{1}{2}$$

$$P\{\hat{x} = \hat{x}_2\} = P(\lambda < x < \infty) = \int_{\lambda}^{\infty} 2e^{-2x}u(x)dx = \frac{1}{2}$$

$$\boxed{Let\ \lambda > 0} \qquad \int_{\lambda}^{\infty} 2e^{-2x}dx = 1/2 = 1/2$$

$$\boxed{\lambda = 0.3465}$$



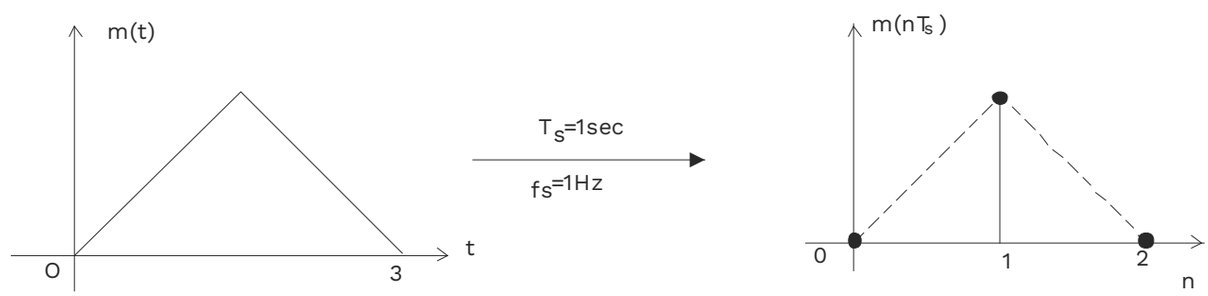
$$\int_0^{\lambda} 2e^{-2x}dx = \int_{\lambda}^{\infty} 2e^{-2x}dx = 1/2 \rightarrow \boxed{\lambda = 0.3465}$$

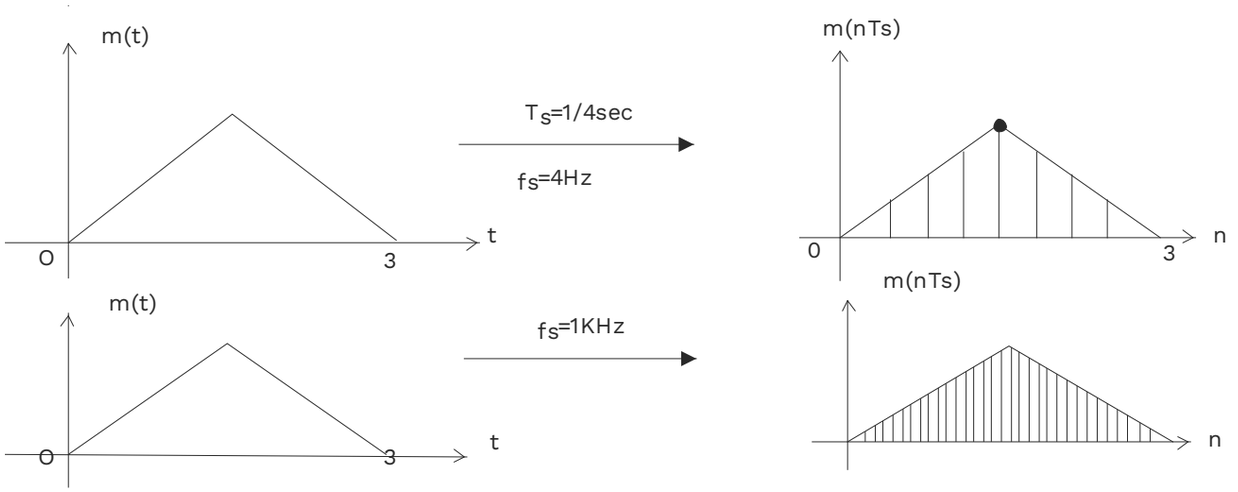
Drawback of PCM

$$Bw = \frac{R_b}{2} = \frac{nf_s}{2} \qquad P_{QE} = \frac{\Delta^2}{12} \qquad \Delta = \frac{DR}{L}$$

$L \le 2^n$

$Bw \downarrow$	$n \downarrow$	$\Delta \uparrow$	$P_{QE} \uparrow$
$P_{QE} \downarrow$	$n \uparrow$	$\Delta \downarrow$	$Bw \uparrow$

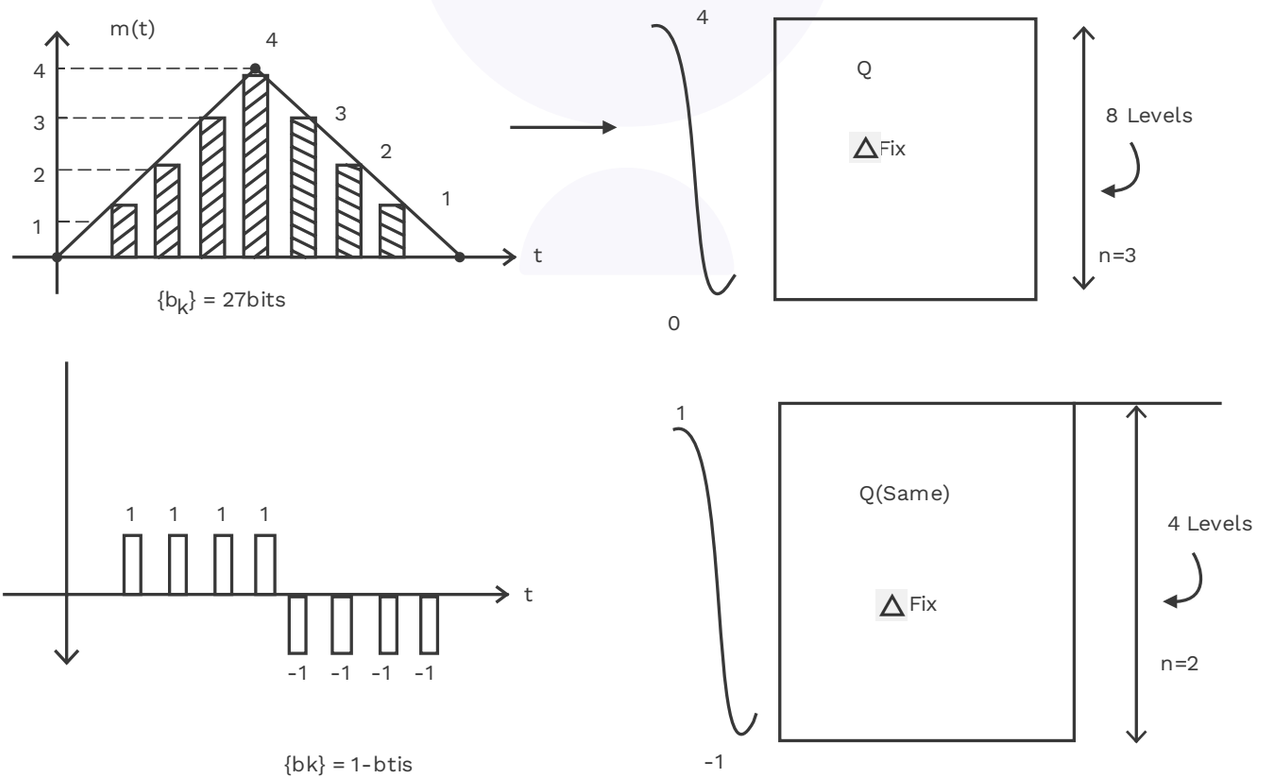




Note:

If a signal is sampled at a rate very well above N.R., then adjacent samples have a very high degree of correlation.

→ It means adjacent samples carry more or less the same information.



Case 2: Has more efficient coding scheme

Or

Less redundant coding scheme

Companding

It is a technique for compressing and then expanding an analog or digital signal. It is used for decreasing noise and cross-talks during transmission. There are two types of companding techniques:

i) μ *law for companding* –

$$y = \frac{1}{\ln(1+\mu)} \ln\left\{1+\mu\left|\frac{m(t)}{m_p}\right| Sgn\left(\frac{m(t)}{m_p}\right)\right\}$$

Where m_p is maximum value of input signal m(t) and μ is constant

$$\begin{cases} \frac{m(t)}{m_p} < 0 & Sgn\left(\frac{m(t)}{m_p}\right) = -1 \\ \frac{m(t)}{m_p} > 0 & Sgn\left(\frac{m(t)}{m_p}\right) = 1 \end{cases}$$ Or

$$y = \frac{1}{ln(1+\mu)} \ln\left(1+\mu\frac{m(t)}{m_p}\right) \quad 0 \le \frac{m(t)}{m_p} \le 1$$

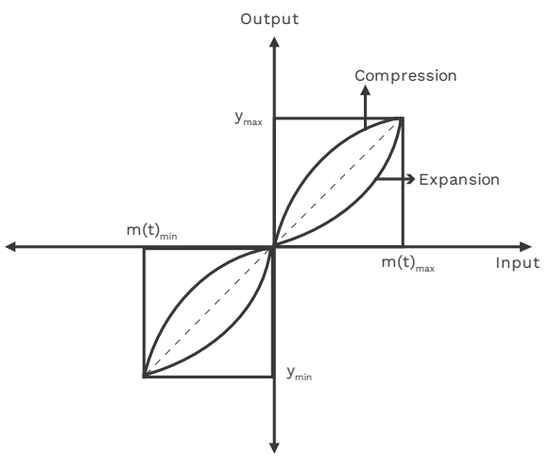


Figure companding

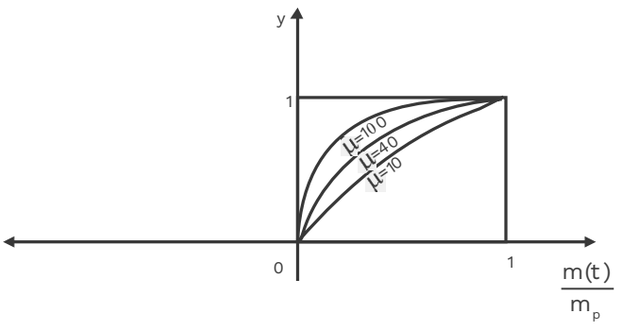


Fig. 5.10 μ **- Law of Companding**

2. A-law for companding:

$$y = \begin{cases} \frac{A}{1+\ln A}\left(\frac{m(t)}{m_p}\right) & 0 \le \frac{m(t)}{m_p} \le \frac{1}{A} \\ \frac{1}{1+\ln A}\left(1+\ln\frac{m(t)}{m_p}\right) & \frac{1}{A} \le \frac{m(t)}{m_p} \le 1 \end{cases}$$

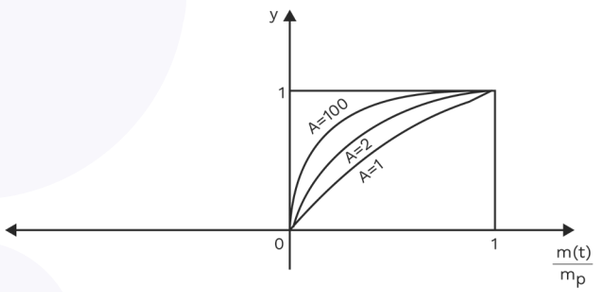


Fig. 5.11 A-law of Companding

For μ law SNR for sinusoidal input is given by:

$$= \frac{3L^2}{\left[\ln(1+\mu)\right]^2} \quad \left[L = \text{no. of levels in Quantizer}\right]$$

Example:

A signal with amplitude range – 6 to 6 is applied to non- uniform with 256 levels. Find bit ratio if signal of B.W 10 KHz is applied to the PCM system.

Solution:

Given

$L = 256$

$L = 2^n$

$2^n = 256$

$n = 8$

Bit rate $R_b = nf_s$

$f_s = 2f_m$

$f_s = 2 \times 10KHz = 20\ KHz$

Now

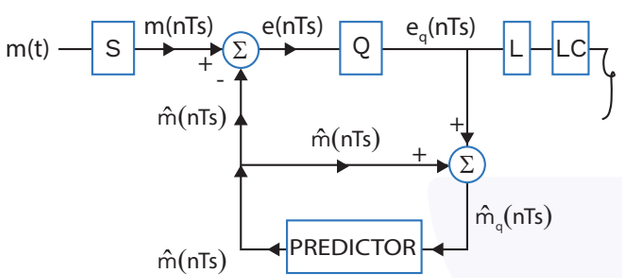
Bit rate

$R_b = 8x\ 20\ KHz$

$R_b = 160KbPs$

Differential Pulse Code Modulation

This technique samples the analog signal and then quantizes the difference between the sampled value and its predicted value. Then encodes the signal to form a digital value.

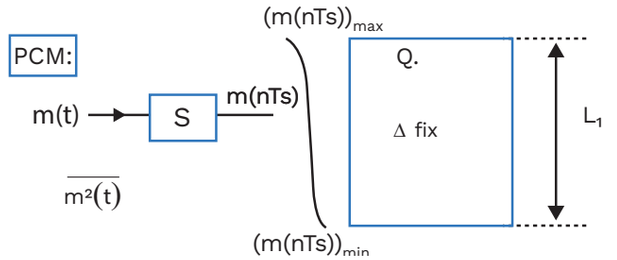


$$e(nTs) = m(nT_s) - \hat{m}(nT_s)$$

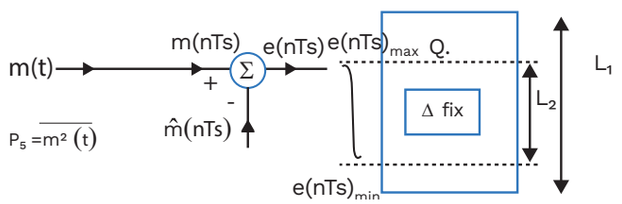
$$m_q(nTs) = \hat{m}(nT_s) + e_q(nT_s)$$

$$QE = e(nTs) - e_q(nT_s)$$

Case 1: Let both PCM and DPCM are Tx are using same Q.



DPCM:



PCM vs DPCM

1. $(P_s)_{PCM} = (P_s)_{DPCM}$
2. $(DR\ at\ Q\ input)_{PCM} > (DR\ at\ Q\ input)_{DPCM}$
3. $\Delta_{PCM} = \Delta_{DPCM} \rightarrow (P_Q)_{PCM} = (P_Q)_{DPCM}$
4. $\Delta = \frac{DR}{L} \rightarrow L_1 > L_2 \rightarrow n_1 > n_2$

   $(n)_{PCM} > (n)_{DPCM} \rightarrow (BW)_{PCM} > (BW)_{DPCM}$
5. $(SQNR)_{PCM} = (SQNR)_{DPCM}$

Case 2: Let PCM and DPCM uses different Q. Having different Δ but same L

1. $(P_s)_{PCM} = (P_s)_{DPCM}$
2. $(DR\ at\ input\ of\ Q)_{PCM} > (DR\ at\ input\ of\ Q)_{DPCM}$
3. $(\Delta)_{PCM} \neq (\Delta)_{DPCM}$ $\qquad \Delta = \frac{DR}{L}$

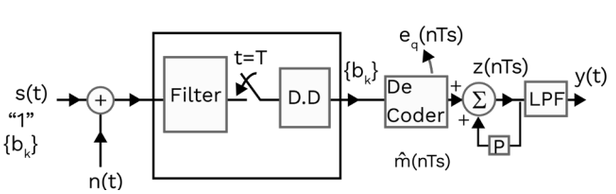
   $\Delta_{PCM} > \Delta_{DPCM}$ $\quad (P_Q)_{PCM} > (P_Q)_{DPCM}$
4. Since L is same

   $n_1 = n_2$ $\qquad (BW)_{PCM} = (BW)_{DPCM}$
5. $(SQNR)_{PCM} < (SQNR)_{DPCM}$

SUMMARY

1. Δ is fix for both Q
   - $(BW)_{PCM} > (BW)_{DPCM}$
   - $(SQNR)_{PCM} = (SQNR)_{DPCM}$
2. L is fix for both Q
   - $(BW)_{PCM} = (BW)_{DPCM}$
   - $(SQNR)_{PCM} < (SQNR)_{DPCM}$
3. In DPCM the difference between present sample and predicted value of present sample is transmitted through wired channel.

DPCM Rx



$$Z(nTs) = e_q(nTs) + \hat{m}(nTs) =$$

$e_q(nTs) + m(nTs) - e(nTs)$

$Z(nTs) = \{m(nTs) - Q.E.\} \rightarrow$

Recovered signal has Q.E.

Drawbacks: Complex Tx and Rx.

Example: Calculate the value of signal to quantization noise ratio if a signal $2\sin\omega t$ is to be encoded with 8 bit PCM with the quantizer having step-size $\Delta = 0.2$.

Solution:

$$SQNR = \frac{Signal\ power}{Quantization\ noise\ power}$$

$$Signal\ power = \frac{A_m^2}{2} = \frac{4}{2} = 2$$

$$Qunatization\ noise\ power = \frac{\Delta^2}{12}$$

$\Delta = 0.2$

$$Qunatization\ noise\ power = \frac{\Delta^2}{12} = \frac{(0.2)^2}{12}$$

$$SQNR = \frac{12}{(0.04)/12} = \frac{24}{0.04}$$

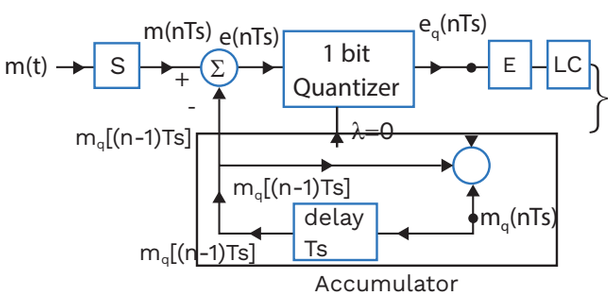
$SQNR = 600$

$(SQNR)_{dB} = 10\log 600 = 27.78\,dB$

Delta Modulation

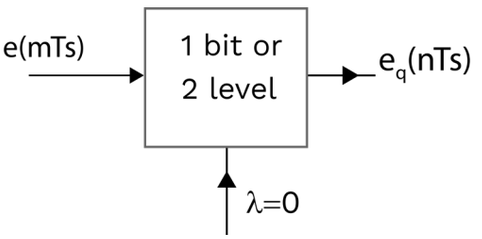
If analog signal is sampled at a very high frequency, then the difference between adjacent samples is very less, which causes a problem in proper recovery of the signal. Now, for better, sampling, we use 1 bit quantizer in the DPCM system.

Tx:



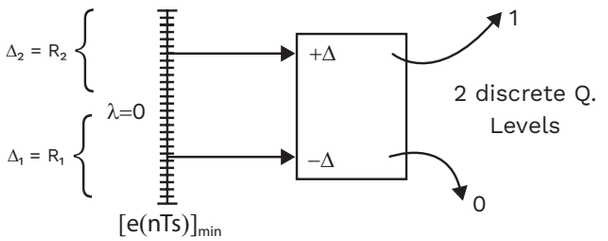
1-bit Quantizer:

1.



$Q.E. = e(nTs) - e_q(nTs)$

2. $[e(nTs)]_{max}$ $Q.output = e_q(nTs)$ *Encoder*



3. $(Q_e)_{max} = \pm\Delta$

4. Let Q.E. = Y → R.V.

$$Y = \begin{cases} defined: & -\Delta \le y \le \Delta \\ 0: & else \end{cases}$$

Let

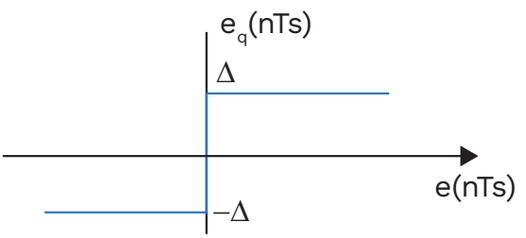
$Y \sim U[-\Delta, \Delta]$

$$P_{QE} = E[X^2 QE] = \frac{\Delta^2}{3}$$

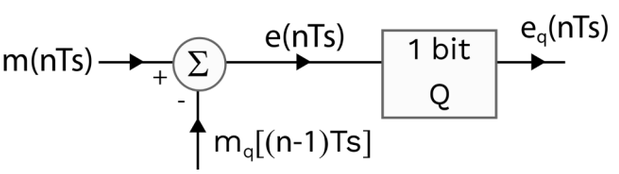
5. $$e_q(nTs) = \begin{cases} +\Delta & e(nTs) > 0 \\ -\Delta & e(nTs) < 0 \end{cases}$$

$e_q(nTs) = +\Delta\,\text{sgn}[e(nTs)]$

6.



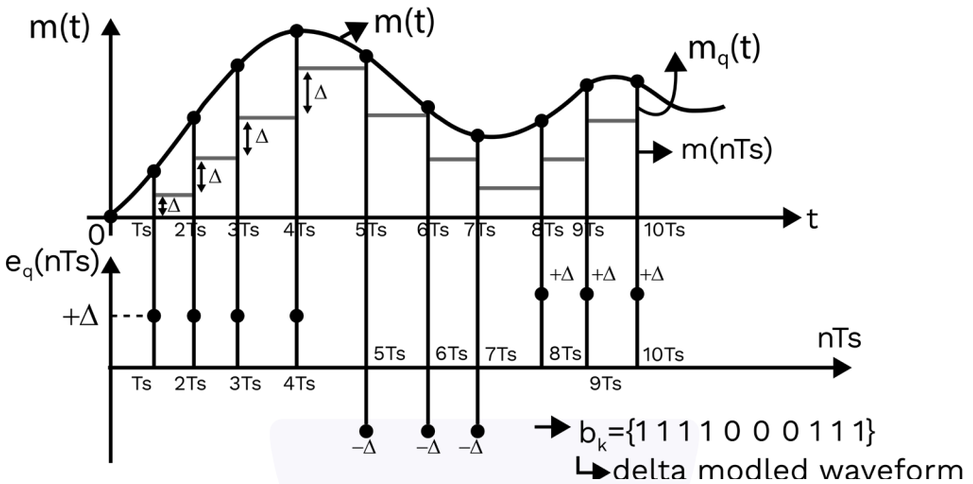
Working:



Let $m(nTs) - m_q\left[(n-1)Ts\right] = m(nTs) - m\left[(n-1)Ts\right]$

$e(nTs) \approx m(nTs) - m\left((n-1)Ts\right)$

$$e_q(nTs) = \begin{cases} +\Delta & e(nTs) > 0 \\ -\Delta & e(nTs) < 0 \end{cases}$$



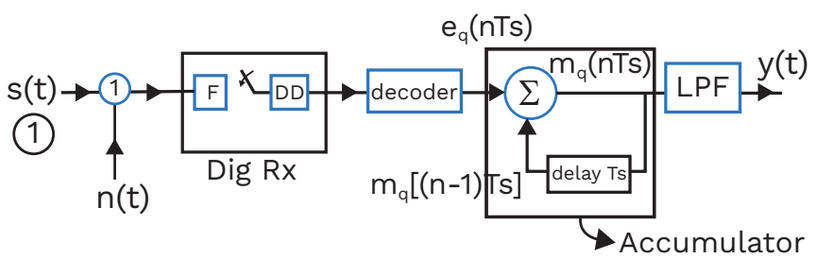
$\rightarrow b_k = \{1111000111\}$

Delta modulated waveform:

$$e_q(nTs) = m(nTs) - m\left[(n-1)Ts\right]$$

$$e_q(Ts) = m(Ts) - m(oTs)$$

Rx:



$m_q(nTs) = e_q(nTs) + m_q(n-1)Ts$

$n = 1 \quad m_q(Ts) = e_q(Ts) + m_q(0Ts) \quad Let\ m_q(0Ts) = 0$

$n = 2 \quad m_q(2Ts) = e_q(2Ts) + m_q(Ts) = e_q(2Ts) + e_q(Ts)$

$n = 3 \quad m_q(3Ts) = e_q(3Ts) + m_q(2Ts) = \sum_{K=1}^{3} e_q(kTs)$

$$m_q(nTs) = \sum_{K=0}^{n} e_q(kTs)$$

$$m_q(nTs) \rightarrow demodulated\ signal$$

Demodulated signal:

$$m_q(nTs) = \sum_{k=0}^{n} e_q(kTs)$$

$$m_q(nTs) = e_q(nTs) + \sum_{k=0}^{n-1} e_q(kTs)$$

$$m_q(nTs) = e_q(nTs) + m_q((n-1)Ts)$$

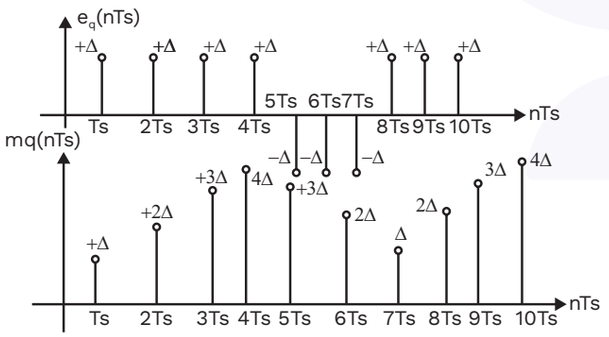
$$e_q(nTs) = m_a(nTs) - m_q((n-1)Ts)$$

$$e_q(nTs) \rightarrow Txed\ signal$$

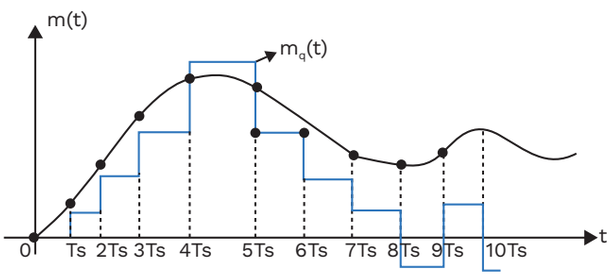
$$m_q(nTs) \rightarrow Rxed\ signal$$

$$e_q(nTs) = m_q(nTs) - m_q(n-1)Ts \rightarrow \underset{\Delta}{I\ difference}$$

Note: $e_q(nTs)$ *and* $m_q(nTs)$ are related by Diff eqⁿ. Hence the name delta modulator.



Tracking of m(t):



Note:

In case of delta modulator-demodulator scheme.

1. The recovered signal is staircase approximation on of Txedanalog message signal.
2. Stairs are added or subtracted at each sampling instance.
3. Size of step in each stair Δ = *step size.*
4. The Rx and Tx have accumulators.

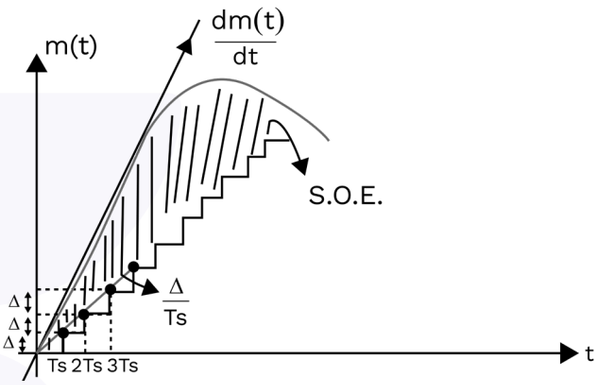
Tracking error in delta modulator:

1. Slope overload error
2. Granular error

Slope overload error:

Δ is very small such that:

$$\left|\frac{dm(t)}{dt}\right|_{max} >>> \frac{\Delta}{Ts}$$

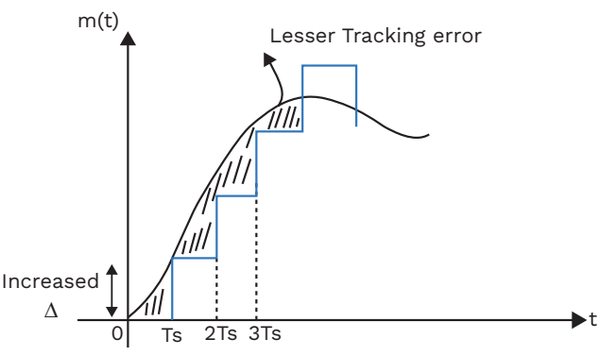


Slope overload error occurs:

$$\left|\frac{dm(t)}{dt}\right|_{max} >>> \frac{\Delta}{Ts}$$

To avoid S.D.E Increase Δ in such a way.

$$\left|\frac{dm(t)}{dt}\right|_{max} \leq \frac{\Delta}{Ts}$$



Case 1: $m(t) = A_m \cos \omega_m t$ or $m(t) = A_m \sin \omega_m t$

Avoid SDE

$$\left|\frac{dm(t)}{dt}\right|_{max} \leq \frac{\Delta}{Ts}$$

$$A_m \omega_m \leq \frac{\Delta}{Ts}$$

$$A_m \leq \frac{\Delta f_s}{2\pi f_m}$$

Case 2: $m(t) = A_{m_1} \cos 2\pi f_{m_1} t + A_{m_1} \cos 2\pi f_{m_2} t$

$$\left|\frac{dm(t)}{dt}\right|_{max} = \left\{A_{m_1} 2\pi f_{m_1} + A_{m_2} 2\pi f_{m_2}\right\}$$

$$A_{m_1} 2\pi f_{m_1} + A_{m_2} 2\pi f_{m_2} \leq \frac{\Delta}{Ts}$$

Example: A sinusoidal signal of 2 kHz is applied to a delta modulator. The sampling rate is 20,000 samples/sec and step size of delta modulator is 0.1 V. To prevent SOE, the maximum amplitude of sinusoidal signal is?

Solution:

$$\left|\frac{dm(t)}{dt}\right|_{max} \leq \frac{\Delta}{T_s}$$

$$A_m \omega_m \leq \frac{\Delta}{Ts}$$

$$A_m \leq \frac{0.1}{\frac{2\pi \times 2 \times 10^3}{2 \times 10^4}}$$

$$A_m \leq \frac{1}{2\pi}$$

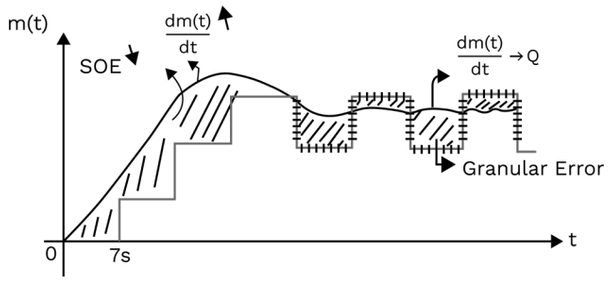
$$(A_m)_{max} = \frac{1}{2\pi} volts.$$

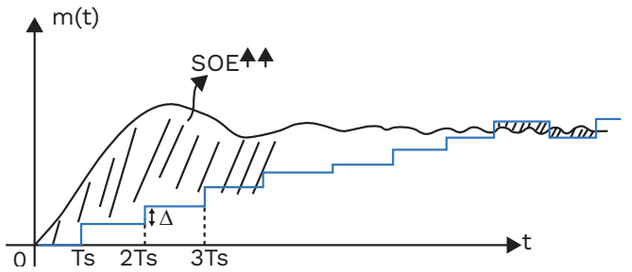
Note: $R_b = nfs \xrightarrow[\Delta \text{modt}^r]{} n = 1$

$R_b = f_s$

Delta modulator: → Bit Rate = Sampling Rate = Pulse Rate.

Granular Error: To avoid SOE $\Delta \uparrow$

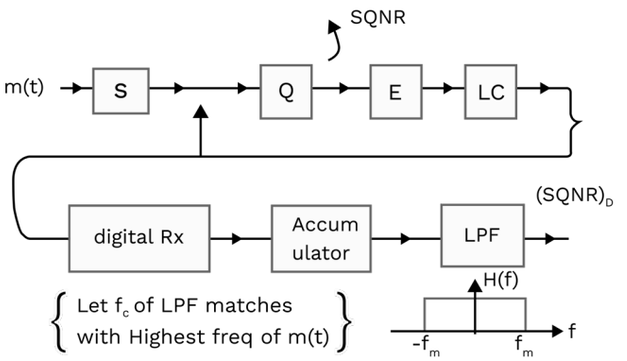




Important note:

1. To remove SOE $\rightarrow \Delta \uparrow \rightarrow$ Leads to increased G.E.
2. To remove G.E. $\rightarrow \Delta \downarrow \rightarrow$ Leads to increased SOE
3. If the recovered signal from delta modt^r Rx is having consecutive pulses of opposite polarity in $t_1 \leq t \leq t_2$, this indicates that during this interval analog signal is essentially constant.

SQNR in delta modt^r:



Let f_c of LPF matches with Highest frequency of m(t)

Case 1: $SQNR = \frac{P_S}{P_{QE}} = \frac{3P_S}{\Delta^2}$

Case 2: $SQNR_D = \frac{P_S}{(P_{QE})_D}$

$$X_{QE} \sim U[-\Delta, \Delta]$$

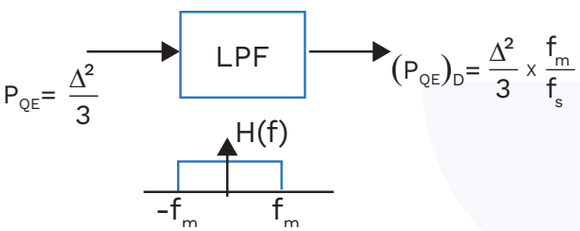
$$S_{QE}(f) = \left(\frac{\Delta^2}{3}\right)\frac{1}{2fs} \rightarrow PSD\ of\ X_{QE}$$



$$\left(S_{XQE}(f)\right)_P = |H(f)|^2 S_{XQE}(f)$$

$$= \begin{cases} \frac{\Delta^2}{6f_s} & -f_m \le f \le f_m \\ 0 & else \end{cases}$$

$$\left(P_{QE}\right)_D = \int_{-\infty}^{\infty}\left(S_{XQE}(f)\right)_D df = \frac{\Delta^2}{3} \times \frac{f_m}{f_s}$$



$$\left(S_{QNR}\right)_D = \frac{3P_S f_s}{\Delta^2 \times f_m} = \left(\frac{3P_S}{\Delta^2}\right) \times \left(\frac{f_s}{f_m}\right)$$

Case 3: SQNR when m(t): Sinusoidal and SOE: Avoided.

$$SQNR = \frac{3P_S}{\Delta^2}$$

$$m(t) = A_m \sin \omega_m t \rightarrow P_s = \frac{A_m^2}{2}$$

$$SOE\ avoid: (\Delta)_{\min} = \frac{A_m \omega_m}{f_s}$$

$$(SQNR)_{\max} = \frac{3 \times Am^2/2}{\frac{A_m^2 \omega_m^2}{f_s^2}} = \frac{3f_s^2}{2\omega_m^2}$$

$$(SQNR)_{\max} = \frac{3}{8\pi^2}\left(\frac{f_s}{f_m}\right)^2$$

Case 4: $(SQNR)_D$ when m(t): Sinusoidal SOE avoided.

$$(SQNR)_{\max} = \frac{3}{80}\frac{f_S^2}{f_m^2}$$

$$\left((SQNR)_D\right)_{\max} = \frac{3}{80}\left(\frac{f_s}{f_m}\right)^3$$

Adaptive delta modulation:

This method is used to solve the granular noise and slope overload error caused during delta modulation. In adaptive delta modulation, the step size of the staircase signal is not fixed and changes depending upon the input signal.

- If output of quantizer continuously changes from 1 to 0 and vice versa, i.e., slope overload distortion. Then step size Δ increases.
- If output of quantizer is 10 10 10 ..... i.e., Granular noise then step size decreases.

M-ary symbolling in PCM:

This is the type of digital modulation technique used for delta transmission in which instead of one bit, two or more bits are transmitted at a time. In this, channel bandwidth is reduced.

Bandwidth of PCM:

	Binary symbolling	*M – ary symbolling*
Rectangular	R_b	$\frac{R_b}{\log_2 M}$
Nyquist	$\frac{R_b}{2}$	$\frac{R_b/2}{\log_2 M}$

$M = 2^n$

where n = Number of bits

And M = Number of combinations, conditions, or levels, with n bits.

Problem in M-ary symbolling:

1. Energy requirement of system will increase.
2. Probability of error will also increase.

Example: If bandwidth required to transmit a binary PCM signal is 1000 kHz, then the bandwidth required for 16-ary PCM is _____ kHz.

Solution:

Given,

Bandwidth for binary PCM = 100 kHz

Bandwidth of binary PCM = $\frac{R_b}{2} = 100\,kHz$

Bandwidth of M-ary PCM = $\frac{R_b / 2}{\log_2 M}$

Bandwidth of 16-ary PCM = $\frac{100}{4} = 25\,kHz$

Hence, bandwidth require for 16-ary PCM = 25 kHz.

Exercise

1. Consider two real valued signals, x(t) bandlimited to [-500 Hz, 500 Hz] and y(t) bandlimited to [- 1 kHz, 1 kHz]. For z(t) = x(t) y(t), the Nyquist sampling frequency (in kHz) is _____.

Solution: 3

2. The input to a 1-bit quantizer is a random variable X with PDF $f_x(x) = 2e^{-2x}$ for $x \geq 0$ and $f_x(x) = 0$ for $x < 0$. For outputs to be of equal probability, the quantizer threshold should be_____.

Solution: 0.346

3. In a PCM system, the signal $m(t) = \sin(100\pi t) + \cos(100\pi t)$ V is sampled at the Nyquist rate. The samples are processed by a uniform quantizer with step size 0.75 V. The minimum data rate of the PCM system in bits per second is _____.

Solution: 200

4. An analog voltage in the range 0 to 8 V is divided in 16 equal intervals for conversion to 4-bit digital output. The maximum quantization error (in V) is _____.

Solution: 0.25

5. A sinusoidal signal of 2 kHz frequency is applied to a delta modulator. The sampling rate and step-size Δ of the delta modulator are 20,000 samples per second and 0.1 V, respectively. To prevent slope overload, the maximum amplitude of the sinusoidal signal (in volts) is

(A) $\frac{1}{2\pi}$

(B) $\frac{1}{\pi}$

(C) 0

(D) None

Solution: A

6. A sinusoidal signal of amplitude A is quantized by a uniform quantizer. Assume that the signal utilizes all the representation levels of the quantizer. If the signal to quantization noise ratio is 31.8 dB, the number of levels in the quantizer is _____.

Solution: 32

7. The signal $x(t) = \sin\left(14000\pi t\right)$, where t is in seconds, is sampled at a rate of 9000 samples per second. The sample signal is the input to an ideal low pass filter with frequency response $H(f)$ as follows:

$$H(f) = \begin{cases} 1, & |f| \leq 12kHz \\ 0, & |f| > 12kHz \end{cases}$$

What are the number of sinusoids in the output and their frequencies in kHz?

(A) Number of sinusoids = 1, Frequency = 7

(B) Number of sinusoids = 3, Frequencies = 2, 7, 11

(C) Number of sinusoids = 2, Frequencies = 2, 7

(D) Number of sinusoids = 2, Frequencies = 7, 11

Solution: B

8. A 4 kHz sinusoidal message signal having amplitude 4 V is fed to a Delta Modulator (DM) operating at a sampling rate of 32 kHz. The minimum step size required to avoid slope overload noise in the DM (rounded off to two decimal places) is _____V.

Solution: 3.14

Solutions

1. Given:

x(t) has bandwidth $f_1 = 500\,\text{Hz}$ and y(t) has bandwidth $f_2 = 1\text{Hz}$.

$z(t) = x(t)y(t)$

Two signal x(t) and y(t) are multiplied in time domain then they will get convolved in frequency domain.

If spectrum of any signal $p(t)$ is defined as,

$$P(f) = \begin{cases} k_1, & -L_1 \le f \le L_1 \\ 0, & \text{otherwise} \end{cases}$$

And spectrum of any other signal $q(t)$ is defined as,

$$Q(f) = \begin{cases} k_2, & -L_2 \le f \le L_2 \\ 0, & \text{otherwise} \end{cases}$$

The spectrum of convolved signal can be written as

$$P(f) \otimes Q(f) = \begin{cases} k, & -(L_1 + L_2) \le f \le (L_1 + L_2) \\ 0, & \text{otherwise} \end{cases}$$

Hence, the spectrum Z(f) of multiplied signal z(t) will have bandwidth

$(L_1 + L_2) = (500 + 1000) = 1500Hz$,

Sampling frequency is given by,

$f_s = 2f_m$

$f_s = 2 \times 1500 = 3kHz$

Hence, the Nyquist sampling frequency is 3 kHz.

2. Given:

At output, probabilities are equal. Assume quantization voltage is V_{th}.

For output of quantizer to have equal probabilities

$P(x \le V_{th}) = P(x \ge V_{th})$

$$\int_{-\infty}^{V_{th}} f_X.(x)dx = \int_{V_{th}}^{\infty} f_X(x)dx$$

$$\int_{0}^{V_{th}} 2e^{-2x}dx = \int_{V_{th}}^{\infty} 2e^{-2x}dx$$

$$\frac{2[e^{-2x}]_0^{V_{th}}}{(-2)} = \frac{2[e^{-2x}]_{V_{th}}^{\infty}}{(-2)}$$

$$-(e^{-2V_{th}} - 1) = -(0 - e^{-2V_{th}})$$

$$1 = 2e^{-2V_{th}}$$

$$V_{th} = 0.346$$

Hence, the quantizer threshold is 0.346.

3. Given:

$S = 0.75V,\ f_m = 50Hz$

$m(t) = \sin(100\pi t) + \cos(100\pi t)$

$$m(t) = \sqrt{(1)^2 + (1)^2}\cos\left[100\pi t - \tan^{-1}\left(\frac{1}{1}\right)\right]$$

From trigonometry identity,

$$\left[a\cos\theta + b\sin\theta = \sqrt{a^2 + b^2}\cos\left(\theta - \tan^{-1}\frac{b}{a}\right)\right]$$

$$m(t) = \sqrt{2}\cos\left[100\pi t - 45^0\right]$$

Step size is given by,

$$S = \frac{V_H - V_L}{2^n} = \frac{\sqrt{2} - (-\sqrt{2})}{2^n}$$

$$0.75 = \frac{2\sqrt{2}}{2^n}$$

$$2^n = 3.77$$

$$L = 3.77 \approx 4$$

To represent four quantization levels at least 2 bits are required. So

$n = 2$

Bits rate is given by,

$$R_b = nf_s$$

Sampling frequency is given by,

$$f_s = 2f_m = 2 \times 50 = 100Hz$$

$$R_b = 2 \times 100 = 200 \text{ bits/sec.}$$

Hence, the minimum data rate of the PCM system is 200 bits / sec.

4. Given:

Voltage range = 0 to 8 V and n = 4

The maximum quantization error is given by,

$$Q_e = \frac{S}{2} \qquad ...(i)$$

Where, S = step size

Step size is given by,

$$S = \frac{V_H - V_L}{2^n} = \frac{8-0}{2^4} = \frac{8}{16} = 0.5$$

From equation (i)

$$Q_e = \frac{0.5}{2} = 0.25$$

Hence, the maximum quantization error is 0.25.

5. Given:

$f_m = 2kHz,\ f_s = 20kHz \text{ and } \Delta = 0.1V$

To avoid slope overload,

$$\frac{\Delta}{T_s} \geq \left|\frac{d}{dt}m(t)\right|_{\max} \qquad ...(i)$$

For single – tone sinusoidal modulating signal,

$$m(t) = A_m \sin \omega_m t$$

$$\frac{d}{dt}m(t) = A_m \omega_m \cos \omega_m t$$

$$\left|\frac{d}{dt}m(t)\right|_{\max} = A_m \omega_m$$

From equation (i)

$$\Delta f_s \geq A_m \omega_m \qquad \left[f_s = \frac{1}{T_s}\right]$$

$$0.1 \times 20 \times 10^3 \geq A_m \times 2\pi \times 2 \times 10^3$$

$$A_m \leq \frac{1}{2\pi}$$

The maximum amplitude of the sinusoidal signal is $\frac{1}{2\pi}$.

Hence, the correct option is (A).

6. Given:

SQNR = 31.8 dB

For sinusoidal signal, signal to quantization noise ratio is given by,

$$(SQNR)_{dB} = 1.76 + 6n$$

$$31.8 = 1.76 + 6n$$

$$6n = 30$$

$$n = 5$$

The relationship between number of quantization levels and number of bits is given by,

$$L = 2^n = 2^5 = 32$$

Hence, the number of levels in the quantizer is 32.

7. **Given:**

$x(t) = \sin(14000\pi t)$,

$f_s = 9000$ samples/sec, $f_m = 7000\,Hz$

Cut – off frequency of LPF = 12kHz.

The frequency components present in sampled signal are,

$$f_m, f \pm f_m, 2f_s \pm f_m, 3f_s \pm f_m .....$$

$$f_m = 7kHz$$

$$f_s \pm f_m = 9 \pm 7 = 16kHz \text{ and } 2kHz$$

$$2f_s \pm f_s = 18 \pm 7 = 25kHz \text{ and } 11kHz$$

$$3f_s \pm f_s = 27 \pm 7 = 34kHz \text{ and } 20kHz$$

The number of components present at the output of LPF will be 3 and they are 2

kHz, 7 kHz and 11 kHz.

Hence, the correct option is (B).

8. Given:

Message signal frequency, $f_m = 4kHz$

Message signal amplitude, $A_m = 4V$

Sampling rate, $f_s = 32kHz$

Assume massage signal is

$m(t) = A_m \cos 2\pi f_m t$

Slop of $m(t) = \dfrac{d}{dt} m(t) = 2\pi f_m A_m \cos 2\pi f_m t$

Maximum slope of $m(t)$ is given by

$$\frac{d}{dt} m(t)\Big|_{max} = 2\pi f_m A_m$$

[$\because$ Maximum value of $(\cos 2\pi f_m t) = 1$]

To avoid slope overload distortion,

$$\Delta . f_s \geq \frac{d}{dt} m(t)\Big|_{max} \text{ where, } \Delta = \text{Step Size.}$$

$$\Delta . f_s \geq 2\pi f_m A_m$$

$$\Delta \geq \frac{2\pi f_m A_m}{f_s}$$

So, minimum value of step size (Δ) is given by,

$$\Delta_{min} = \frac{2\pi f_m A_m}{f_s}$$

$$\Delta_{min} = \frac{2\pi \times 4 \times 4}{32} = \pi = 3.14V$$

Hence, the minimum step size required to avoid slope overload noise in the DM is 3.14 V.